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 - Media Storage: Cloudinary
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Feature Desk G - Complete Documentation

1. Project Overview

WHAT IS FEATURE DESK?

A revolutionary smart study desk with:

-  **Flip-top surface** (Normal desk ↔ Digital touchscreen)
-  **Pressure-sensitive stylus pen** for natural writing
-  **Integrated AI-powered educational platform** (10 core AI capabilities)
-  **Integrated Analytics Dashboard** (Student + Teacher + School)
-  **Virtual Labs** (Chemistry, Physics, Biology)

SUMMARY: WHY FEATURE DESK IS UNIQUE

NO DIRECT COMPETITOR EXISTS for Feature Desk's combination of:

Unique Selling Point	Description
 Flip-Top Smart Desk	First desk that transforms from normal surface to digital touchscreen
 Natural Writing Experience	Stylus pen with pressure sensitivity on desk surface
 10 Core AI Capabilities	Gemini-powered adaptive learning, grading, and analytics
 Complete Educational Ecosystem	Student + Teacher + School Admin/Analytics in one platform
 Virtual Labs	50 interactive PhET simulations across Physics/Chemistry/Biology with 3D Three.js interface

Unique Selling Point	Description
💰 Affordable AI	AI token costs as low as ₹50-100/month for 1,000 students (see full TCO in §7.4)
IN India-First Design	Built for CBSE/ICSE/State Board K-12 curriculum

2. System Architecture

2.1 Technology Stack

- **Frontend:** React (v18.3), TypeScript (v5.5), Vite (v7.3)
- **Styling:** Tailwind CSS (v3.4), Lucide React (v0.344, Icons), Framer Motion (v12.23)
- **AI Integration (Dual-Brain Strategy):**
 - **Gemini 2.5 Flash:** Primary model for complex reasoning, content generation, grading, and multimodal analysis (~\$0.50/month per 1000 students).
 - **Gemini 2.5 Flash Lite:** Cost-efficient model for high-frequency tasks like auto-naming, text extraction, and micro-analytics (~\$0.10/month per 1000 students).
 - **API Key Fallback System:** Multiple Gemini API keys configured with automatic rotation on rate-limit errors ([gemini.ts](#)). If one key is exhausted, the system seamlessly falls back to the next key.
- **Databases (Triple-Layer Architecture):**
 - **Supabase (PostgreSQL):** *Structured Data & Relations*. Source of Truth for Authentication, User Roles, Class Schedules, Results, Assessments, Question Banks, and Real-time Notifications. JSONB columns store complex question data.
 - **Firebase (Firestore):** *Real-time Chat & Feedback Data*. Specialized for storing **AI Chatbot Sessions**, **Chat History**, and **Detailed Student Feedback** (question-wise breakdowns).
 - **Cloudinary (Cloud Media):** *Media Files & Images*. Handles all file uploads — teacher PDFs/documents, exam answer sheet images, canvas JSON data, and profile images via unsigned upload preset.
- **PDF Processing:** [pdfjs-dist](#) (v5.4) for client-side PDF text extraction
- **Charts & Visualization:** Recharts (v3.2) for analytics dashboards
- **Export:** html2canvas (v1.4) + jsPDF (v4.1) for PDF generation
- **Spreadsheet:** xlsx (v0.18) for batch question import (CSV/Excel parsing)
- **3D:** Three.js (v0.160) + React Three Fiber for Virtual Labs
- **State Management:** React Context API

- **Deployment:** Netlify (with `netlify.toml` configuration)

2.2 Database Roles Detailed

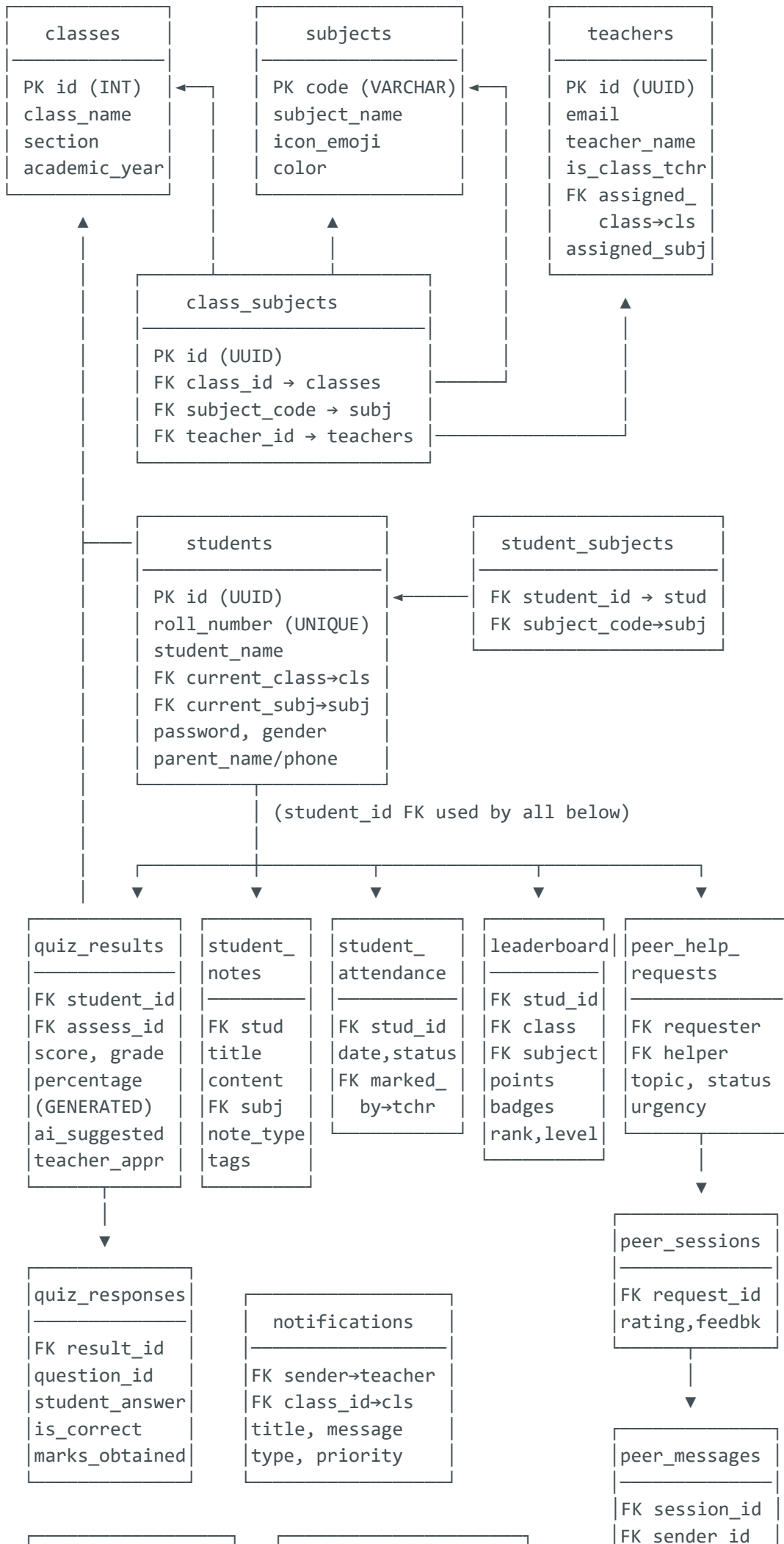
- **Supabase (PostgreSQL):** Handles relational integrity (Student → Class → Subject). Real-time subscriptions power the live collaboration and notification features. JSONB columns store assessment questions and detailed quiz responses. Auto-calculated percentage fields in `quiz_results`. Row Level Security (RLS) policies on sensitive tables.
- **Firebase (Firestore):** Handles AI chatbot conversation persistence and **Detailed Exam Feedback**. The `firestoreService.ts` manages:
 - **Chat:** `createChatSession()`, `saveChatMessage()`, `subscribeToChat()` for real-time study assistance.
 - **Feedback:** `saveStudentFeedback()` stores rich, JSON-heavy grade reports in the `student_feedback` collection, including question-wise marks, AI suggestions, and teacher comments that would otherwise bloat the SQL database.
- **Cloudinary:** Handles all media uploads via unsigned preset (`feature_desk_unsigned`). The `cloudinaryService.ts` auto-detects `resource_type` (image/raw/video/auto) based on MIME type. Supports `uploadFile()`, `uploadPdf()`, `uploadDocument()`, and `uploadJson()`. Used in exam submission (answer sheet images), canvas notes (JSON blobs), and teacher content uploads.

2.3 Database Implementation Matrix (Supabase + Firebase + Cloudinary)

Feature / Module	Database Implementation	Implementation Details
User Authentication	Supabase	Secure login for Students (roll+password), Teachers (email+password), Admins
Student Profiles	Supabase	<code>students</code> table with 20+ fields (roll, name, class, parent details, health)
Class/Subject Logic	Supabase	<code>classes</code> , <code>subjects</code> , <code>class_subjects</code> , <code>student_subjects</code> tables

Feature / Module	Database Implementation	Implementation Details
Assessments	Supabase	<code>assessments</code> table with JSONB <code>questions</code> column, <code>exam_type</code> classification
Quiz Results & Grades	Supabase	<code>quiz_results</code> table with auto-calculated <code>percentage</code> (GENERATED ALWAYS AS)
Detailed Responses	Supabase	<code>quiz_responses</code> table (per-question answers, marks, <code>time_spent</code>)
Question Banks	Supabase	<code>study_materials</code> table with JSONB for AI-generated questions
Chatbot Sessions	Firebase Firestore	<code>chat_sessions</code> collection containing <code>userId</code> , <code>title</code> , <code>lastMessageAt</code>
Chat History	Firebase Firestore	<code>chat_history</code> collection with session-aware message storage
Detailed Feedback	Firebase Firestore	<code>student_feedback</code> collection for question-wise grading breakdowns
File Uploads (PDF/Images)	Cloudinary	<code>cloudinaryService.uploadFile()</code> with auto <code>resource_type</code> detection
Exam Answer Sheets	Cloudinary + Supabase	Images → Cloudinary, metadata/grades → Supabase <code>quiz_results</code>
Canvas Data (Strokes)	Cloudinary + Supabase	JSON blobs → Cloudinary, metadata → Supabase <code>student_notes</code>
Notifications	Supabase	<code>notifications</code> table with Real-time subscriptions
Attendance	Supabase	<code>student_attendance</code> table with daily tracking
Leaderboard	Supabase	<code>leaderboard</code> table with points, badges, streaks, levels
Peer Help System	Supabase	<code>peer_help_requests</code> , <code>peer_sessions</code> , <code>peer_messages</code> tables

2.4 Entity-Relationship (ER) Diagram



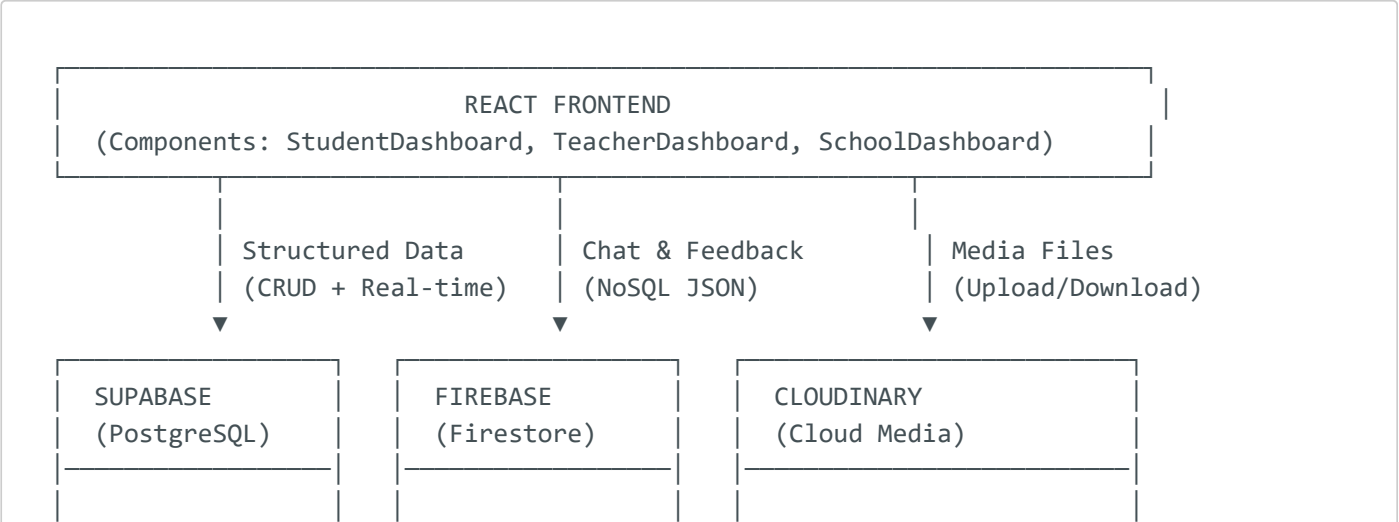
study_materials	student_interactions	content, type
FK subject→subj	FK student_id → stud	
FK class→cls	FK subject→subj	
FK uploaded_by→tc	interaction_type	
file_url, ai_sum	details (JSONB)	
key_topics, tags	duration	

assessments
PK id (UUID)
FK subject_code → subjects
FK class_id → classes
FK created_by → teachers
title, exam_type, questions (JSONB), total_marks
time_limit, passing_marks, shuffle_questions
instructions, scheduled_at, is_active

Key Relationships:

- **classes** ← **students** (many students per class)
- **subjects** ← **assessments**, **student_notes**, **leaderboard** (many records per subject)
- **teachers** → **assessments** (created_by), **notifications** (sender), **study_materials** (uploaded_by)
- **students** → **quiz_results**, **student_notes**, **student_attendance**, **leaderboard**, **peer_help_requests**
- **quiz_results** → **quiz_responses** (detailed per-question breakdown)
- **peer_help_requests** → **peer_sessions** → **peer_messages** (chat chain)

2.5 Triple-Layer Data Flow Diagram



- Students
- Teachers
- Classes/Subj
- Assessments
- Quiz Results
- Quiz Responses
- Notifications
- Attendance
- Leaderboard
- Peer System
- Study Mats
- Interactions

RLS Policies ✓
Real-time ✓
JSONB Columns ✓

- chat_sessions
- chat_history
- student_feedbk

subscriptions
for AI chatbot

- Exam answer sheets
(image/auto)
- Teacher PDFs
(raw)
- Canvas JSON data
(raw/json)
- Documents (Word/Excel)
(raw)

Upload Preset:
feature_desk_unsigned

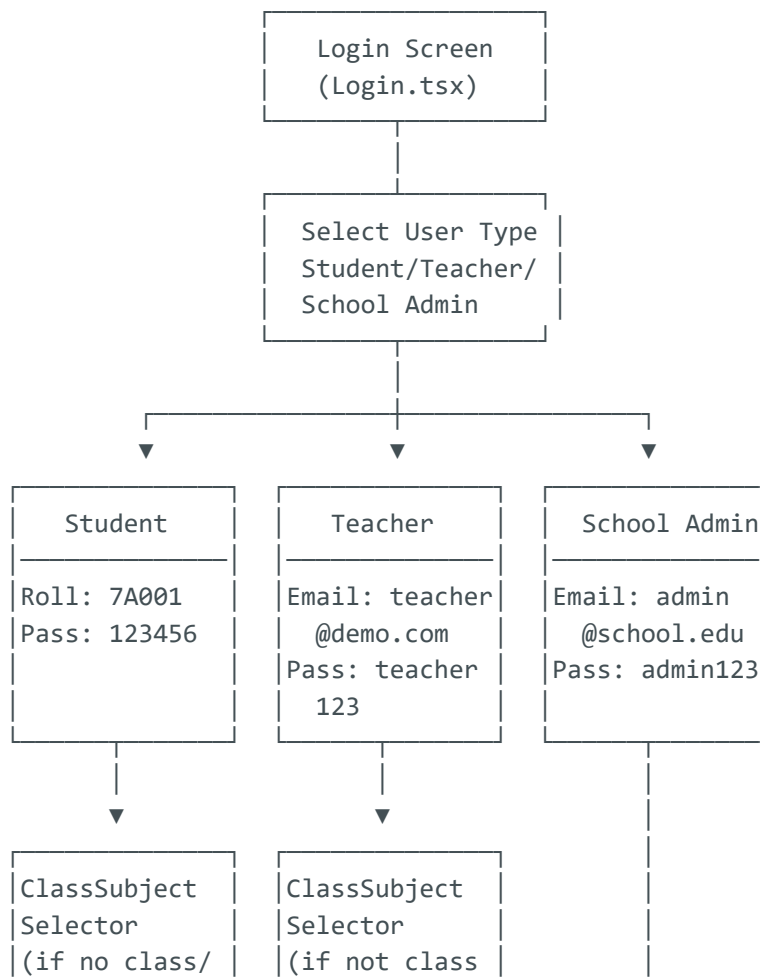
Auto resource_type:
image→auto, pdf→raw
doc→raw, json→raw

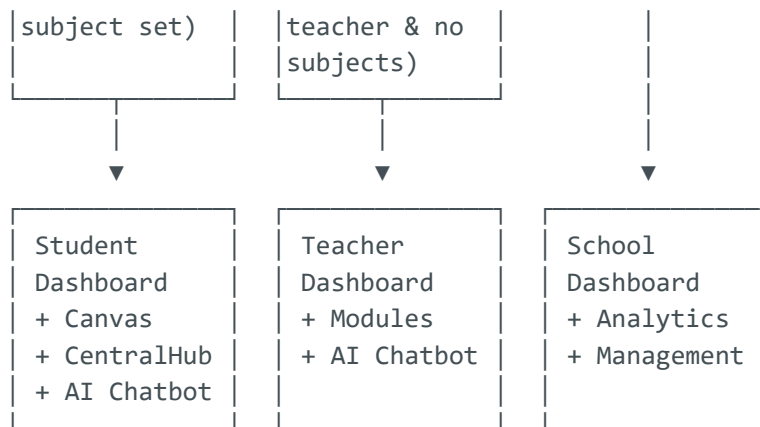
Service Files:
supabase.ts
teacherDb.ts
schoolDb.ts
questionDb.ts
db.ts (hybrid helper)

Service File:
firebaseService.ts
firebase.ts

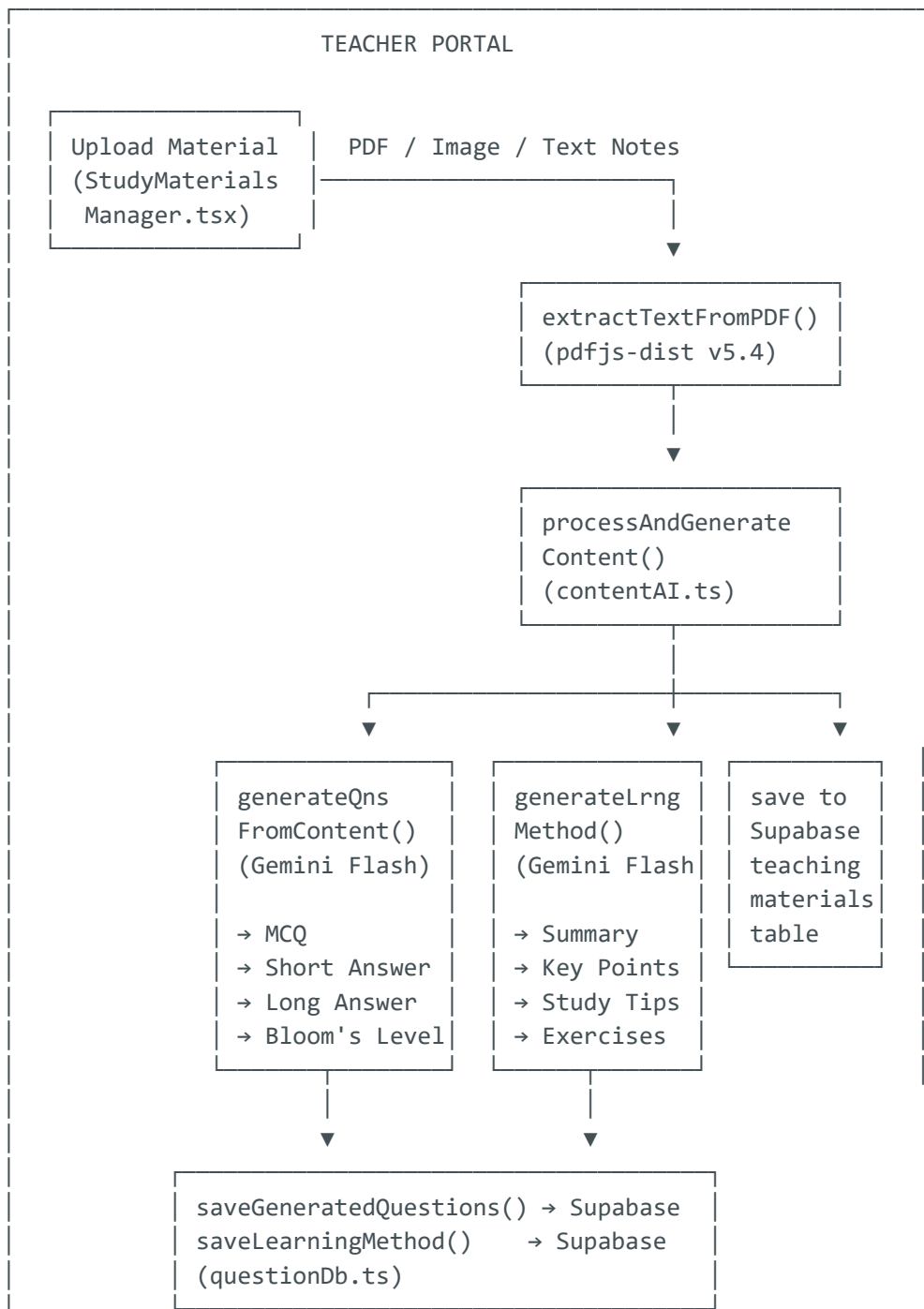
Service File:
cloudinaryService.ts

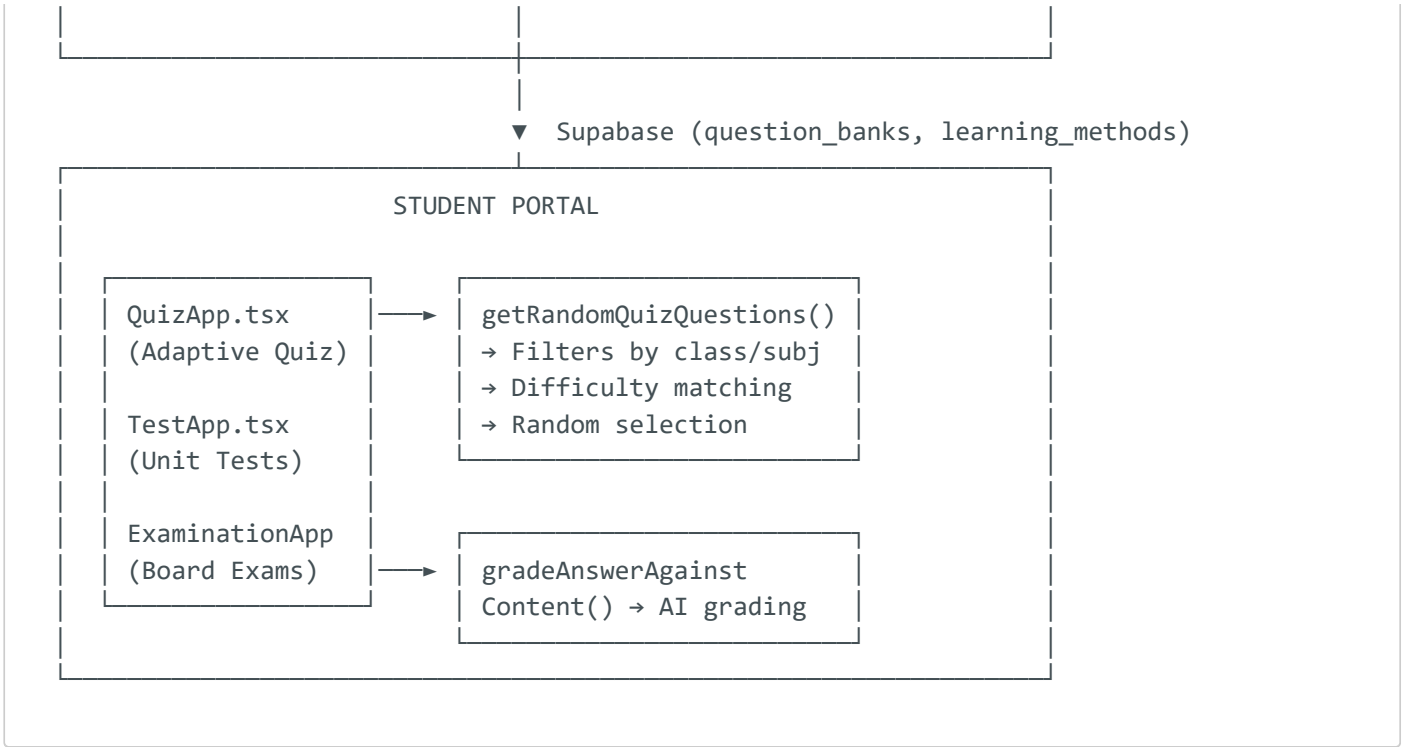
2.6 Authentication Flow Diagram





2.7 AI Content Pipeline Flow Diagram





3. Feature Breakdown

3.1 Student Portal - Detailed Application Breakdown

Core Interface Components

Writing Canvas ([WritingCanvas.tsx](#) - 67KB)

A distraction-free, infinite digital notebook that serves as the primary workspace for students.

Implementation Details:

- **Canvas Technology:** HTML5 Canvas with custom stroke rendering
- **Pressure Simulation:** Velocity-based stroke width calculation for natural pen feel
- **Infinite Canvas:** Virtual scrolling with viewport-based rendering
- **Paper Styles:** Lined, grid, blank, and graph paper templates
- **Smart Toolbar:**
 - Pen types: Ballpoint, Fountain, Highlighter, Marker
 - Eraser modes: Stroke eraser, Point eraser, Area eraser
 - Color palette with recent colors

- Undo/Redo with unlimited history
- **Storage:** All stroke data serialized to **Cloudinary** (JSON blobs) with metadata in **Supabase** for replay capability

Central Hub (**CentralHub.tsx** - 12KB)

A floating "Super Animation" button that expands to reveal 12 integrated applications.

Implementation Details:

- **Radial Menu:** 360° expansion with smooth elastic animation
- **Drag & Drop:** Hub can be repositioned anywhere on screen
- **App Cards:** Each app shows icon, name, description, and preview image
- **Navigation:** React Router integration for seamless app switching
- **Memory:** Saves last position in localStorage

 Student Applications

#	App Name	Component	Size	Primary Function
1	Quiz App	QuizApp.tsx	29KB	Adaptive daily practice
2	Unit Test App	TestApp.tsx	20KB	Timed chapter assessments
3	Examination App	ExaminationApp.tsx	25KB	High-stakes board-style exams
4	Dashboard Analysis	DashboardAnalysis.tsx	15KB	Performance self-reflection
5	Live Chatbot	LiveChatbot.tsx	11KB	AI study companion
6	Floating AI Chatbot	FloatingAIChatbot.tsx	21KB	Context-aware assistant
7	Notes App	NotesApp.tsx	14KB	Digital notebook with AI naming
8	History Viewer	HistoryViewer.tsx	12KB	Activity timeline
9	Notification Center	NotificationCenter.tsx	10KB	Teacher communications

#	App Name	Component	Size	Primary Function
10	Life Activity App	LifeActivityApp.tsx	11KB	Ethical reasoning scenarios
11	Social Learning	SocialLearningDashboard.tsx	29KB	Peer collaboration
12	Virtual Labs	Route:/virtual-labs	-	Unity WebGL experiments

App 1: Quiz App (Adaptive Learning)

File: QuizApp.tsx | Size: 29KB

Purpose: Daily practice with AI-powered adaptive difficulty adjustment.

Key Features:

- **Teacher Content Integration:** Pulls questions from teacher-uploaded PDFs
- **AI Fallback:** Generates questions via Gemini when no teacher content available
- **Socratic Hints:** Progressive hint system (Conceptual → Formula → Step-by-step)
- **Immediate Feedback:** Detailed explanations for every wrong answer
- **Adaptive Reinforcement:** Generates targeted follow-up questions for weak areas

Implementation Flow:

```
1. checkTeacherContent() → Verify if teacher uploaded materials
2. generateQuiz() → Pull from Supabase question bank
3. generateWithAI() → Fallback to Gemini API
4. handleAnswerSelect() → Record answer + response time
5. generateSocraticHints() → AI hint generation on wrong answers
6. handleNextQuestion() → Adapt difficulty based on performance
```

Data Model:

```
interface QuizQuestion {
  id: number | string;
  question: string;
  options: string[];
  correct: number;
  explanation: string;
  timeEstimate: number;
  marks?: number;
  difficulty?: 'easy' | 'medium' | 'hard';
}
```

```
sourceContentTitle?: string; // Links to teacher's uploaded PDF
}
```

App 2: Unit Test App (Chapter Assessments)

File: `TestApp.tsx` | Size: 20KB

Purpose: Timed chapter-wise assessments with basic lockdown features.

Key Features:

- **Subject-wise Organization:** Tests grouped by Mathematics, Science, English, etc.
- **Search & Filter:** Find tests by subject or keywords
- **Timed Assessment:** Configurable duration per test (25-40 minutes typical)
- **Score History:** View completed test results
- **Question Navigation:** Jump to any question during the test

Implementation Flow:

1. Display available tests (mockTests data)
2. `handleStartTest(test)` → Initialize timer, lock navigation
3. `handleSelectAnswer()` → Store answers in local state
4. `handleNextQuestion()` → Navigate with boundary checks
5. `calculateScore()` → Compute final score on submission

Test Structure:

```
interface Test {
  id: string;
  title: string;           // "Algebra Unit Test"
  subject: string;         // "Mathematics"
  duration: number;        // Minutes
  questions: TestQuestion[];
  completed?: boolean;
  score?: number;
}
```

App 3: Examination App (High-Stakes Testing)

File: `ExaminationApp.tsx` | Size: 25KB

Purpose: Board exam simulation with full lockdown security features.

Key Features:

- **Secure Environment:**
 - Tab-switching detection with warning count
 - Fullscreen enforcement
 - Copy-paste disabled
 - Right-click blocked
- **Question Types:**
 - Multiple Choice (MCQ)
 - Short Answer (100-200 words)
 - Essay/Long Answer (500+ words)
 - File Upload (for diagrams/images)
- **Progress Tracking:** Visual question grid showing answered/unanswered
- **Auto-save:** Saves progress every 2 minutes to prevent data loss
- **Time Warnings:** Alerts at 15min, 5min, 1min remaining

Security Implementation:

```
// Tab switch detection
document.addEventListener('visibilitychange', handleVisibilityChange);

// Fullscreen enforcement
const toggleFullScreen = () => {
  if (!document.fullscreenElement) {
    document.documentElement.requestFullscreen();
  }
};
```

Exam Data Model:

```
interface Exam {
  id: string;
  title: string;
  subject: string;
  duration: number;           // Total minutes
  totalMarks: number;
  instructions: string;      // Pre-exam instructions
  questions: Question[];
  startTime: string;         // ISO timestamp
  endTime: string;
  status: 'upcoming' | 'active' | 'completed';
}
```

File: `DashboardAnalysis.tsx` | **Size:** 15KB

Purpose: Self-reflection on learning progress with AI-generated insights.

Key Features:

- **Visual Analytics:**
 - Subject-wise performance bar charts
 - Weekly progress line graphs
 - Strength/weakness pie charts
 - **AI Insights:**
 - "Am I Ready?" readiness check
 - Personalized study recommendations
 - Confidence calibration analysis
 - **Goal Tracking:** Set and monitor learning goals
 - **Streak Counter:** Days of continuous study
-

App 5 & 6: AI Chatbots

Files: `LiveChatbot.tsx` (11KB), `FloatingAIChatbot.tsx` (21KB)

Purpose: Context-aware AI study companion available anytime.

Key Features:

- **Context Awareness:** Knows current page content, recent quiz performance
- **Multilingual:** Supports Hindi, English, and regional languages
- **Subject Expertise:** Tailored responses for Math, Science, Languages
- **Conversation History:** Continues previous discussions
- **Floating Access:** Always-available button in bottom corner

Gemini Integration:

```
const response = await generateChatResponse({
  message: userMessage,
  context: {
    currentPage: canvasContext,
    recentPerformance: lastQuizResults,
    studentProfile: learningStyle
  }
});
```

App 7: Notes App (Digital Notebook)

File: `NotesApp.tsx` | Size: 14KB

Purpose: Organized digital notebook with AI-powered features.

Key Features:

- **Auto-Naming:** AI analyzes handwriting to suggest page titles
 - **Smart Templates:** Pre-built templates for Lab Reports, Essays, Summaries
 - **Folders:** Organize by Subject, Date, or Custom categories
 - **Search:** Full-text search across all notes
 - **Export:** PDF/Image export for sharing
-

App 8: History Viewer (Activity Timeline)

File: `HistoryViewer.tsx` | Size: 12KB

Purpose: Complete timeline of all learning activities.

Key Features:

- **Activity Types:** Quizzes, Tests, Notes, Canvas sessions
 - **Date Filtering:** View by day, week, month
 - **Time Spent:** Track study hours per subject
 - **Replay:** Access any past canvas session
-

App 9: Notification Center

File: `NotificationCenter.tsx` | Size: 10KB

Purpose: Centralized hub for all teacher communications.

Key Features:

- **Real-time Updates:** Supabase real-time subscriptions
 - **Categories:** Assignments, Announcements, Reminders, Feedback
 - **Priority Levels:** Urgent, Normal, Low
 - **Mark as Read:** Track read/unread status
-

App 10: Life Activity App (Ethical Reasoning)

File: LifeActivityApp.tsx | **Size:** 11KB

Purpose: Scenario-based ethical decision-making exercises.

Key Features:

- **Real-world Scenarios:** Age-appropriate ethical dilemmas
 - **AI Consequences:** Dynamic outcome generation based on choices
 - **Reasoning Feedback:** AI critiques the logic behind decisions
 - **Character Development:** Tracks ethical reasoning growth
-

App 11: Social Learning Dashboard

File: SocialLearningDashboard.tsx | **Size:** 29KB

Purpose: Peer collaboration and group study features.

Key Features:

- **Peer Chat:** Real-time messaging with classmates
 - **Study Groups:** Create/join topic-focused groups
 - **Leaderboards:** Friendly competition with weekly rankings
 - **Collaborative Whiteboard:** Multi-user canvas sessions
 - **Peer Tutoring:** Request help from high-performing peers
-

App 12: Virtual Labs (Science Hub)

Technology: Three.js 3D Engine + PhET HTML5 Simulations + Sharded Netlify Deployment

Live URL: <https://scilab-hubx.netlify.app>

Integration: Accessed from Student Portal via CentralHub → opens external Science Hub platform

Architecture:

- **3D Lab Interface:** Three.js-powered carousel with UnrealBloomPass glow effects, starfield backgrounds, and dynamic card textures generated via HTML5 Canvas
- **Universal Experiment Wrapper:** `experiment-wrapper.html` loads all simulations in an `<iframe>` with consistent UI (back button, loading animation, title)

- **Central Registry:** `js/experiments.js` — single source of truth for all 50 experiments (IDs, titles, URLs, categories, lab assignments)
- **Sharded Deployment:** Split across **23 Netlify sub-sites** (`scilab-sims1` through `scilab-sims23`) to handle PhET simulation size limits, with automated `deploy-unique.ps1` PowerShell deployment
- **Extensible:** Supports ANY web-based simulation — PhET, Falstad Circuit, ChemCollective, or custom HTML5 labs

Available Labs (50 Experiments):

Lab	Count	Categories	Example Experiments
Physics	~27	Mechanics, Energy, Electricity, Waves, Optics, Gravity, Fluids, Thermo	Forces & Motion, Circuit Builder, DC/AC, Ohm's Law, Pendulum Lab, Projectile Motion, Energy Skate Park, Bending Light, Wave on a String
Chemistry	~13	Solutions, Reactions, Atomic, Gas, Matter	pH Scale, Acid-Base Solutions, Balancing Equations, Build a Molecule, Build an Atom, Molarity, Beer's Law Lab
Biology	~10	Genetics, Cellular, Neuroscience, Perception, Environment	Natural Selection, Gene Expression, Membrane Transport, Neuron Simulation, Greenhouse Effect

Navigation Flow:

```

Student CentralHub → "Virtual Lab" card → scilab-hubx.netlify.app
  → Select Lab (Physics / Chemistry / Biology)
    → 3D Carousel of Experiment Cards
      → Click "Launch" → experiment-wrapper.html?id=[SIM_ID]
        → iframe loads PhET simulation from scilab-simsX.netlify.app
          → "Back to Lab" returns to carousel
  
```

Key Files (Separate Repository):

File	Purpose
<code>index.html</code>	Main landing page — lab selection
<code>physics-lab.html</code>	Physics 3D carousel (Three.js)
<code>chemistry-lab.html</code>	Chemistry 3D carousel

File	Purpose
biology-lab.html	Biology 3D carousel
experiment-wrapper.html	Universal simulation launcher (iframe)
js/experiments.js	Master database of all 50 experiments
deploy-unique.ps1	Automated multi-site Netlify deployment

App 13: Assessment Launcher (**AssessmentLauncher.tsx** - 13KB)

Purpose: Unified starting point for all quizzes, tests, and exams.

Features:

- **Content Source Selection:** Choose between Teacher-assigned content or AI-generated practice
 - **Pre-Flight Checks:** Verifies internet connection and fullscreen capability
 - **Mode Selection:** Classic, Time-Attack, or Zenith Mode (Hard)
 - **Topic Selection:** Choose specific chapters/topics for AI generation
-

App 14: Handwriting Converter (**HandwritingConverter.tsx** - 6KB)

Technology: Gemini 2.5 Flash Vision

Functionality:

- **Screenshot & Convert:** Takes a snapshot of the current canvas
 - **AI OCR:** Converts handwritten notes to editable digital text
 - **Export:** Copy to clipboard or save as **.txt** file
 - **Smart Formatting:** Preserves bullet points and lists
-

App 15: Peer Review System (**PeerReview.tsx** - 32KB)

Purpose: Anonymous peer feedback mechanism for subjective answers.

Workflow:

1. Student submits subjective answer
2. System anonymizes the answer
3. Answer is distributed to 3 random peers
4. Peers rate (1-5 stars) and provide constructive feedback

- 5. AI moderates feedback for politeness
- 6. Student receives aggregated peer feedback + AI summary

⚡ **App 16: Adaptive Quick-Quiz** (AdaptiveQuiz.tsx - 7KB)

Purpose: Rapid-fire micro-learning sessions.

Key Logic:

- **3-Minute Sessions:** Designed for short attention spans
- **Dynamic Difficulty:**
 - Correct answer → Harder next question (+20 XP)
 - Wrong answer → Easier next question + Hint (+5 XP)
- **Streak Multiplier:** Consecutive correct answers boost XP gain based on streak length

AI Capabilities Used (Student Portal)

The Student Portal leverages **7 of 10** core AI capabilities (see Section 3.4 for full details):

#	AI Capability	Student-Specific Usage
2	 Socratic Hint Engine	Escalating hints on wrong answers, concept explanations
4	 Learning Analytics Engine	Reaction time analysis, confidence insights, study readiness checks
5	 Context-Aware AI Chatbot	Page-aware study companion with conversation history
6	 Handwriting Intelligence	Auto-page naming, screenshot graph analysis, handwriting OCR
7	 Auto-Generated Communications	Weekly narratives, motivation messages
8	 Adaptive Learning Loop	Targeted practice questions for weak areas
9	 Ethical Reasoning Engine	Scenario consequences, reasoning feedback

3.2 Teacher Portal - Detailed Application Breakdown

📊 Core Interface Components

Teacher Dashboard (`TeacherDashboard.tsx` - 19KB)

The main control center for teachers with quick access to all functions.

Dashboard Stats:

- Total Students under management
- Active Quizzes/Assessments count
- Average Class Performance percentage
- Pending Grading queue count
- Students Needing Intervention alerts

Navigation Tabs (10 Tabs):

Tab	Icon	Component	Description
Overview		Built-in	Quick stats, recent activities, AI features showcase
Analytics		<code>AnalyticsDashboard</code>	Deep class performance insights with charts
Assessments		<code>AssessmentManager</code>	CRUD for all quizzes, tests, and exams
Students		<code>StudentManagement</code>	Student profiles, passwords, intervention
Content		<code>ContentManager</code>	AI-powered content upload and tagging
Materials		<code>StudyMaterialsManager</code>	File upload to Cloudinary, AI summaries
Grading		<code>GradingCenter</code>	AI-assisted grading with teacher approval
Import		<code>BatchQuestionImport</code>	Bulk Excel/CSV question upload
Collaborate		<code>TeacherCollaboration</code>	Shared question banks ecosystem

Tab	Icon	Component	Description
Broadcast		<code>NotificationBroadcaster</code>	Send targeted notifications to students

Quick Actions (Always Visible):

-  Create Quiz button → Opens `AssessmentCreator` modal
-  Notification bell → Opens `NotificationBroadcaster` modal
-  Refresh button → Reloads all dashboard data

 Teacher Portal Roles

Role	Permissions	Typical User
Class Teacher	Student profiles, Password resets, Broadcast communications, Attendance	Homeroom Teacher
Subject Teacher	Content upload, Assessment creation, Grading, Analytics	Math/Science/English Teacher
Department Head	All Subject Teacher permissions + Cross-class analytics	Senior Faculty

 Unit Test Creator (Separate Module)

Location: Teacher Portal → Assessment Creator → Unit Tests Tab

Purpose: Create chapter-wise assessments for regular classroom testing.

Key Features:

- **Duration:** 20-45 minutes typical
- **Question Types:** MCQ only (auto-graded)
- **Lockdown:** Basic (no fullscreen enforcement)
- **Scheduling:** Assign to classes with due dates
- **AI Generation:** Generate from uploaded chapter PDFs

Implementation Flow:

1. Select Subject & Chapter
2. Upload PDF/Textbook content OR

3. Use existing question bank
4. AI generates questions with difficulty classification
5. Teacher reviews & modifies
6. Set time limit, marks, due date
7. Assign to one or multiple classes

Unit Test Creation Interface:

```
interface UnitTestConfig {
    title: string;
    subjectCode: string;
    chapterId: number;
    duration: number;           // 20-45 minutes
    questionCount: number;     // 10-25 questions
    totalMarks: number;
    difficultyMix: {
        easy: number;          // percentage
        medium: number;
        hard: number;
    };
    dueDate: string;
    assignedClasses: number[];
}
```



Exam Creator (Separate Module)

Location: Teacher Portal → Assessment Creator → Examinations Tab

Purpose: Create formal examinations (Mid-term, Final, Board-style).

Key Features:

- **Duration:** 1-3 hours typical
- **Question Types:** MCQ, Short Answer, Long Answer, File Upload
- **Lockdown:** Full security (tab detection, fullscreen, copy-paste block)
- **Proctoring:** Activity logging
- **Rubric Creation:** Define grading criteria for subjective answers
- **Scheduling:** Specific date & time with exam hall assignment

Implementation Flow:

1. Select Exam Type (Mid-term, Final, Board Practice)
2. Create Question Paper structure (Sections A, B, C)
3. Add questions manually OR AI-generate from syllabus
4. For subjective questions, define rubric criteria

5. Set strict timing (start & end datetime)
6. Enable security features
7. Preview as student would see
8. Publish to assigned classes

Exam Creation Interface:

```
interface ExamConfig {
  title: string;
  type: 'midterm' | 'final' | 'board_practice' | 'quarterly';
  subjectCode: string;
  duration: number;           // 60-180 minutes
  totalMarks: number;
  sections: ExamSection[];
  securitySettings: {
    fullscreenRequired: boolean;
    tabSwitchLimit: number;
    copyPasteBlocked: boolean;
  };
  scheduledStart: string;    // ISO datetime
  scheduledEnd: string;
  rubrics: RubricCriteria[]; // For subjective grading
}

interface ExamSection {
  name: string;              // "Section A - MCQ"
  instructions: string;
  marks: number;
  questions: Question[];
}

interface RubricCriteria {
  criterion: string;         // "Logical Flow"
  maxPoints: number;
  levels: {
    excellent: string;      // "Clear thesis, supporting arguments"
    good: string;
    average: string;
    poor: string;
  };
}
```



Comparison: Unit Test vs Examination Creation

Feature	Unit Test Creator	Exam Creator
Typical Duration	20-45 minutes	1-3 hours
Question Types	MCQ only	MCQ, Short, Long, File Upload
Grading	100% Auto-graded	MCQ auto, Subjective AI-assisted
Security Level	Basic (timer, no back-navigation)	Full lockdown (tab detection, fullscreen)
Scheduling	Due date (flexible)	Exact time slot (strict)
Rubrics	Not required	Required for subjective
Frequency	Weekly/Bi-weekly	Monthly/Quarterly
AI Features	Question generation only	Generation + Grading + Feedback
Results	Immediate	After teacher approval

Analytics Dashboard ([AnalyticsDashboard.tsx](#) - 24KB)

Purpose: Deep insights into class performance and learning patterns.

Visualization Components (Using Recharts library):

- **Bar Charts:** Subject-wise average scores
- **Line Charts:** Performance trends over time
- **Pie Charts:** Difficulty distribution of assessments
- **Heatmaps:** Topic-wise strength/weakness matrix

AI-Powered Insights:

- **Mistake Pattern Detection:** "60% of class confused velocity with speed"
- **Readiness Forecast:** "Class 8A is 72% prepared for upcoming algebra test"
- **Learning Velocity Narrative:** "Priya improved 15% faster than class average"

Student Management ([StudentManagement.tsx](#) - 21KB)

Purpose: Complete student profile and academic management.

Features:

- **Student Profiles:** View/edit personal details

- **Password Management:** Reset student passwords
 - **Academic Records:** View all assessment history
 - **Intervention Tracking:** Flag and monitor struggling students
 - **Parent Contact:** Integrated communication logs
 - **Attendance Integration:** Sync with school attendance system
-

Content Manager (**ContentManager.tsx** - 24KB)

Purpose: Upload, organize, and manage teaching materials.

Supported Content Types:

- **PDFs:** Textbook chapters, Notes, Worksheets
- **Images:** Diagrams, Charts, Handwritten problems
- **Videos:** Lecture recordings, Explainer videos
- **Links:** External resources, YouTube videos

AI Integration:

- **Auto-Tagging:** AI analyzes content and suggests tags
 - **Question Extraction:** Identifies questions from uploaded materials
 - **Difficulty Classification:** Auto-rates extracted questions
-

Notification Broadcaster (**NotificationBroadcaster.tsx** - 14KB)

Purpose: Send targeted communications to students and parents.

Notification Types:

- **Assignment Reminders:** Upcoming due dates
 - **Announcements:** Class/school-wide messages
 - **Individual Feedback:** Personalized messages to students
 - **Parent Updates:** Weekly progress summaries
 - **Intervention Alerts:** Compassionate messages for struggling students
-

Grading Center (**GradingCenter.tsx** - 23KB)

Purpose: AI-assisted grading with teacher approval workflow.

Workflow:

1. Student submits assessment
2. MCQs auto-graded immediately
3. Subjective answers → AI grades with rubric
4. AI provides confidence score (0-100%)
5. Teacher reviews AI grade
6. Teacher approves OR modifies
7. Results released to student

AI Grading Features:

- **Rubric-Based Scoring:** Grades against teacher-defined criteria
 - **Partial Credit:** Awards points for correct methods
 - **Explanation Generation:** Creates feedback for each answer
 - **Grade Appeal Handling:** Re-evaluates disputed grades
-

👤 Teacher Collaboration ([TeacherCollaboration.tsx](#) - 43KB)

Purpose: Shared question bank ecosystem for cross-teacher resource sharing.

Key Features:

- **Question Bank Sharing:** Create private, school-wide, or public question sets
 - **Collaborator Roles:** Viewer (read-only), Editor (can modify), Owner (full control)
 - **Starring System:** Star/bookmark favorite question banks for quick access
 - **Invite System:** Send email invites to other teachers to collaborate
 - **Search & Filter:** Find question banks by subject, topic, or popularity
-

📁 Batch Question Import ([BatchQuestionImport.tsx](#) - 25KB)

Purpose: Bulk upload of questions via Excel/CSV templates.

Features:

- **Template Download:** Provides pre-formatted Excel/CSV templates
 - **Validation Engine:** Client-side validation for required fields, options, and correct answers
 - **Error Preview:** Highlights invalid rows with specific error messages before upload
 - **Bulk Creation:** Converts valid rows into Supabase [assessments](#) or [study_materials](#) records
 - **Supported Formats:** [.xlsx](#), [.xls](#), [.csv](#)
-

 Assessment Manager ([AssessmentManager.tsx](#) - 31KB)

Purpose: Central hub for managing existing quizzes, tests, and exams.

Capabilities:

- **CRUD Operations:** Edit settings, delete assessments, duplicate existing ones
- **Status Toggling:** Activate/Deactivate assessments instantly
- **Filtering:** Filter by class, subject, status (active/draft/completed)
- **Stats Preview:** Quick view of submission counts and average scores per assessment
- **Bulk Actions:** Delete multiple old assessments at once

 Study Materials Manager ([StudyMaterialsManager.tsx](#) - 34KB)

Purpose: dedicated file organization system for teaching resources.

Features:

- **File Upload:** Drag-and-drop upload to Cloudinary (PDFs, Images, Docs)
- **Organization:** Tagging system (Unit 1, Algebra, Homework)
- **AI Summary:** Auto-generates 3-line summaries of uploaded PDFs
- **Access Control:** Visible to specific classes or shared with other teachers
- **Preview Mode:** Integrated document viewer for checking content before publishing

AI Capabilities Used (Teacher Portal)

The Teacher Portal leverages **7 of 10** core AI capabilities (see Section 3.4 for full details):

#	AI Capability	Teacher-Specific Usage
1	 Intelligent Question Generation	PDF-to-questions, image-to-quiz, difficulty classification
3	 AI Grading System	Rubric-based grading, partial credit, grade appeals
4	 Learning Analytics Engine	Mistake patterns, readiness forecast, learning velocity
5	 Context-Aware AI Chatbot	Interactive document chat for question creation

#	AI Capability	Teacher-Specific Usage
7	 Auto-Generated Communications	Personalized feedback, parent reports, intervention alerts
8	 Adaptive Learning Loop	Question effectiveness analysis
10	 Smart Content Utilities	Video topic suggestions

3.3 School Analytics & Admin Dashboard

Note: This portal focuses strictly on **Educational Analytics** and **School Performance**. It is NOT an ERP system (no fees/transport/HR). It exists to support the core learning mission.

Technology: React + Supabase (School Role) **Status:** LITE VERSION IMPLEMENTED (Analytics, Attendance, Risk Analysis)

 **Dashboard Overview** ([SchoolDashboard.tsx](#) - 67KB)

Purpose: Central command center for student performance and teacher effectiveness.

 **Dashboard Overview Section**

Key Metrics Display:

- Total Students (Active/Inactive)
- Total Teachers (Present/Absent today)
- Total Classes (Active sessions now)
- School-wide Average Performance
- Today's Attendance Rate

Real-time Alerts:

- Teacher absences requiring substitutes
- Students at dropout risk
- Upcoming important dates (Board exams, PTM)
- Performance anomalies (class trending down)

 **Substitute Management System**

Purpose: Handle teacher absences with AI-generated lesson plans.

Workflow:

1. Teacher marks absence (or admin records)
2. System identifies affected classes
3. AI generates lesson plan for substitute
4. Substitute receives lesson plan + materials
5. Parents notified of temporary change
6. Normal teacher receives catch-up report

AI Lesson Plan Generation:

```
interface LessonPlan {
    absentTeacher: string;
    subject: string;
    class: string;
    period: number;
    generatedPlan: {
        objective: string;
        warmUp: string;           // 5 min activity
        mainLesson: string;       // 25 min content
        activity: string;         // 10 min practice
        closure: string;          // 5 min wrap-up
        homework: string;
        materials: string[];
    };
    previousTopicsCovered: string[];
    nextScheduledTopic: string;
}
```

School Analytics Section

Performance Dashboards:

- **Class-wise Rankings:** Compare performance across sections
- **Subject-wise Trends:** Identify strong/weak subjects school-wide
- **Teacher Effectiveness:** Compare class outcomes (anonymized)
- **Year-over-Year:** Historical performance comparison

AI-Powered Reports:

- **Annual School Report:** Narrative for board meetings
 - **Benchmarking Insights:** Compare vs. district/state averages
 - **Dropout Risk Analysis:** Predict and prevent dropouts
-

Scheduler & Calendar

Features:

- **Master Timetable:** School-wide class schedules
- **Exam Calendar:** Board exam dates, internal exam schedules
- **Event Management:** PTM, Sports Day, Cultural events
- **Holiday Calendar:** Auto-adjust for local holidays
- **Board Exam Countdown:** Syllabus coverage vs. time remaining

Intervention & Risk Management

Dropout Risk Dashboard:

- AI analyzes attendance, grades, engagement patterns
- Risk scores (High/Medium/Low) for each student
- Recommended interventions
- Parent outreach tracking

Academic Intervention:

- Flag underperforming students
- Auto-assign remedial content
- Track improvement over time

AI Capabilities Used (School Portal)

 OPTIONAL / FUTURE DEVELOPMENT

The School Portal leverages **3 of 10** core AI capabilities (see Section 3.4 for full details):

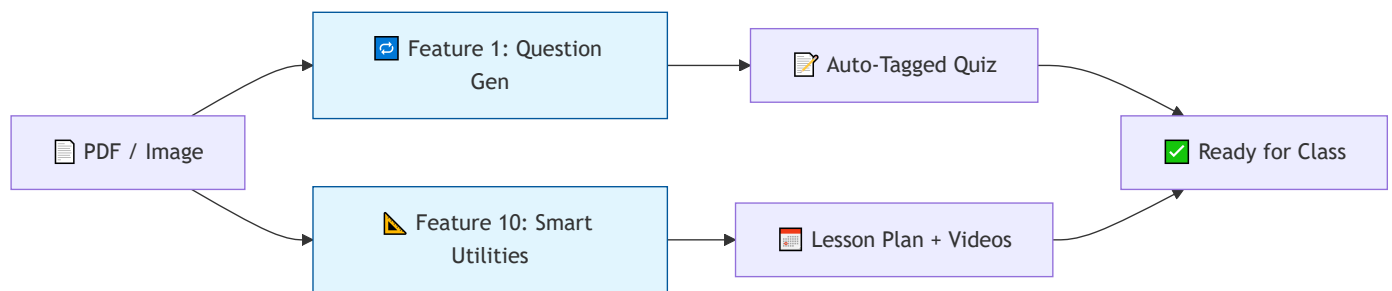
#	AI Capability	School-Specific Usage
4	 Learning Analytics Engine	Dropout risk analysis, benchmarking, board exam readiness
7	 Auto-Generated Communications	Annual school report, absent day notifications
10	 Smart Content Utilities	Auto lesson plans for substitutes

3.4 The AI-Powered Learning Cycle (10 Core Capabilities)

Instead of isolated tools, Feature Desk G uses **10 distinct AI engines** working in a continuous loop to automate the entire education lifecycle.

Phase 1: Teacher Preparation (Zero-Touch Creation)

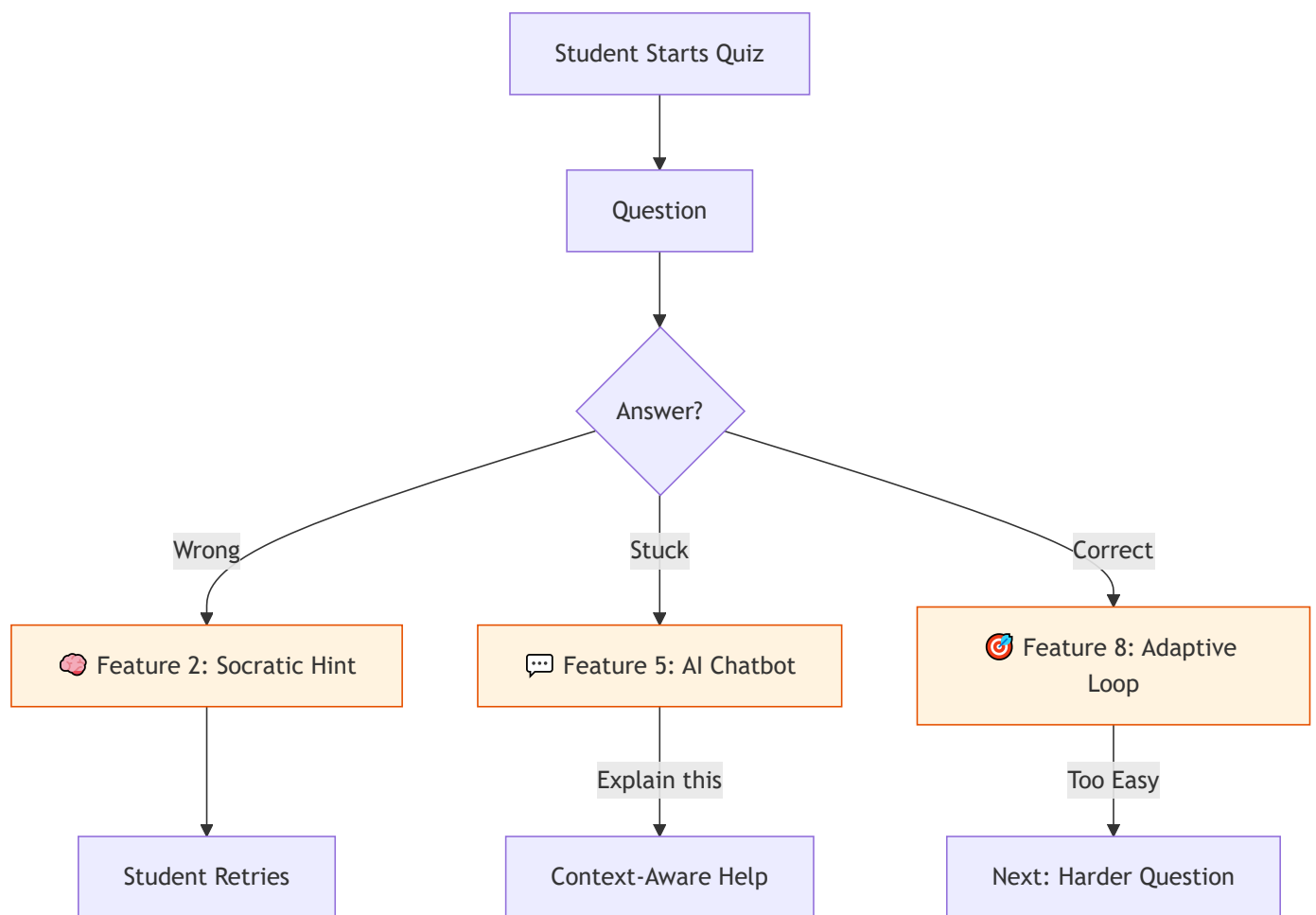
The teacher uploads a PDF chapter. The AI takes over.



1.  **Intelligent Question Generation:** Instantly converts raw content (PDFs, images, text) into Bloom's Taxonomy-tagged assessments (MCQs, Short/Long answers).
2.  **Smart Content Utilities:** Automatically generates lesson plans, prerequisite mind maps, and suggests curated educational videos to supplement the material.

Phase 2: Student Learning (Active & Adaptive)

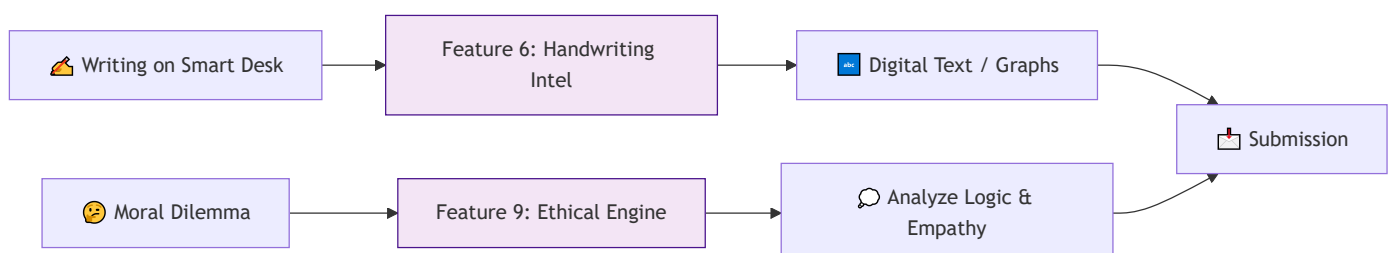
The student struggles with a concept. The AI intervenes.



3. 🧠 **Socratic Hint Engine:** When a student answers incorrectly, it doesn't reveal the answer. It provides escalating hints (Conceptual → Strategic → Step-by-Step) to guide them to the solution.
4. 💬 **Context-Aware AI Chatbot:** Two implementations — `LiveChatbot.tsx` (full-page, session-based, Firebase-backed) and `FloatingAIChatbot.tsx` (overlay that detects current page context — Writing Canvas, Dashboard, Notes — and adjusts suggestions accordingly). Both use Gemini with role-aware prompts (Student vs Teacher).
5. 🎯 **Adaptive Learning Loop:** Dynamically adjusts difficulty based on real-time performance, generating targeted remediation questions for weak concepts.

Phase 3: Assessment & Expression

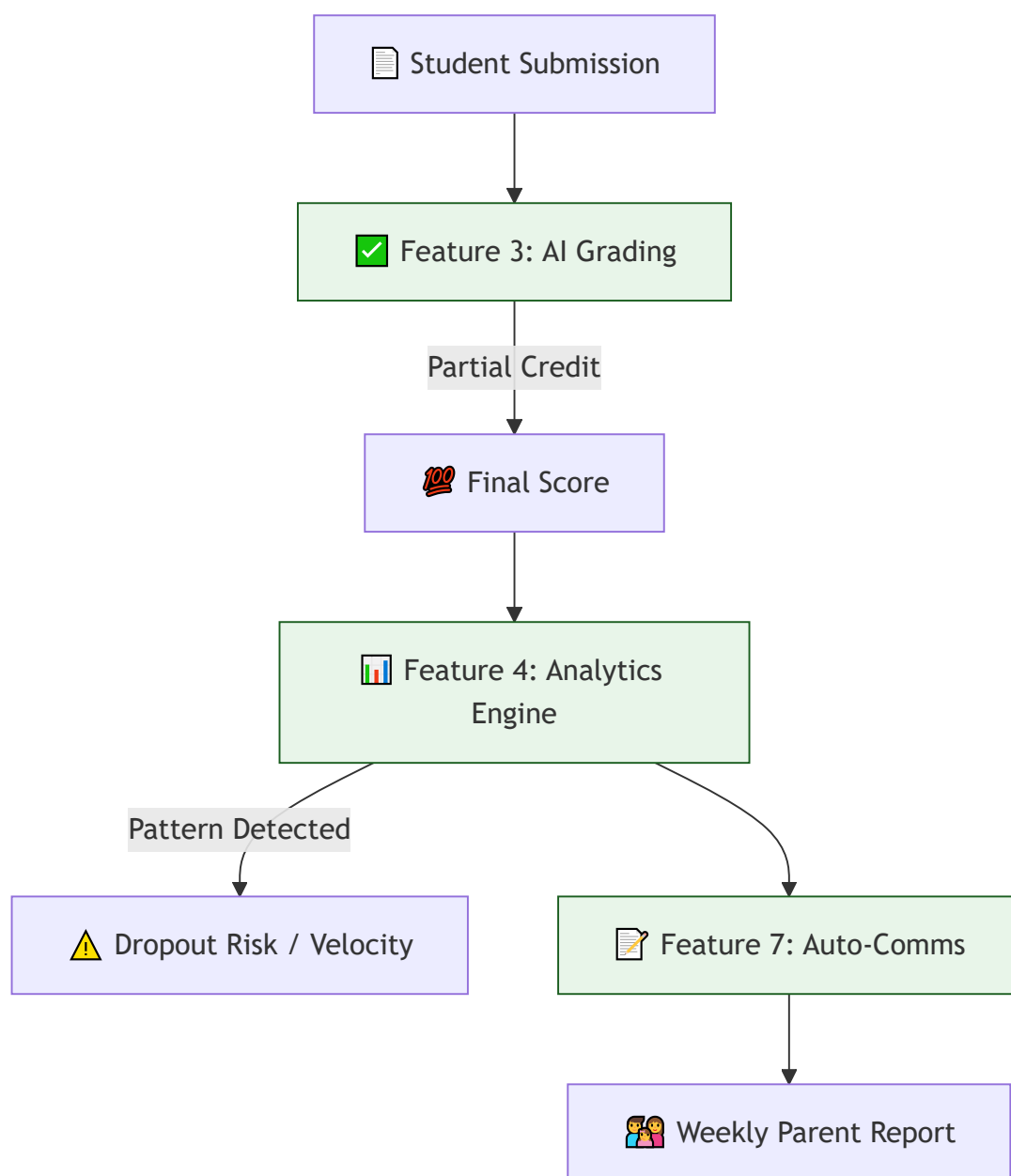
The student takes a high-stakes exam on the Smart Desk.



6.  **Handwriting Intelligence:** The student writes naturally on the desk. The AI recognizes handwriting, converts it to text, and even analyzes drawn diagrams/graphs for accuracy.
7.  **Ethical Reasoning Engine:** For non-STEM subjects, it presents moral dilemmas and critiques the logic and empathy of the student's reasoning, not just the final choice.

Phase 4: Grading & Insight (The Feedback Loop)

The exam is submitted. The teacher gets actionable data, not just scores.



8.  **AI Grading System:** Grades subjective, handwritten answers against a teacher-defined rubric. It awards partial credit for correct methodology and provides a confidence score for teacher review.
9.  **Learning Analytics Engine:** Goes beyond grades to detect "Guessing Patterns," "Learning Velocity," and "Dropout Risk." It tells the teacher *who* needs help and *why*.

10. 📝 **Auto-Generated Communications:** `generatePersonalizedFeedback()` and `generateInterventionMessage()` are **live in UI** (GradingCenter, StudentManagement). `generateParentReport()` and `generateWeeklyNarrative()` are **API-ready** (functions exist in `teacherAI.ts/gemini.ts`, UI integration pending). Annual reports are generated via `schoolAI.ts`.

Unique Enhancements (Cross-Portal)

- **Flow State Mode:** Pomodoro-style distraction blocking
 - **Semantic Knowledge Graph:** Visual node-graph connecting concepts (**Supabase JSONB**)
 - **Collaborative Whiteboard:** Multi-user shared canvas
 - **AI "Ghost" Replay:** Stroke-by-stroke writing replay (**Cloudinary JSON Storage**)
-

4. Parallel Implementation Mapping

 **Teacher Portal ↔ Student Portal**

Feature Sync

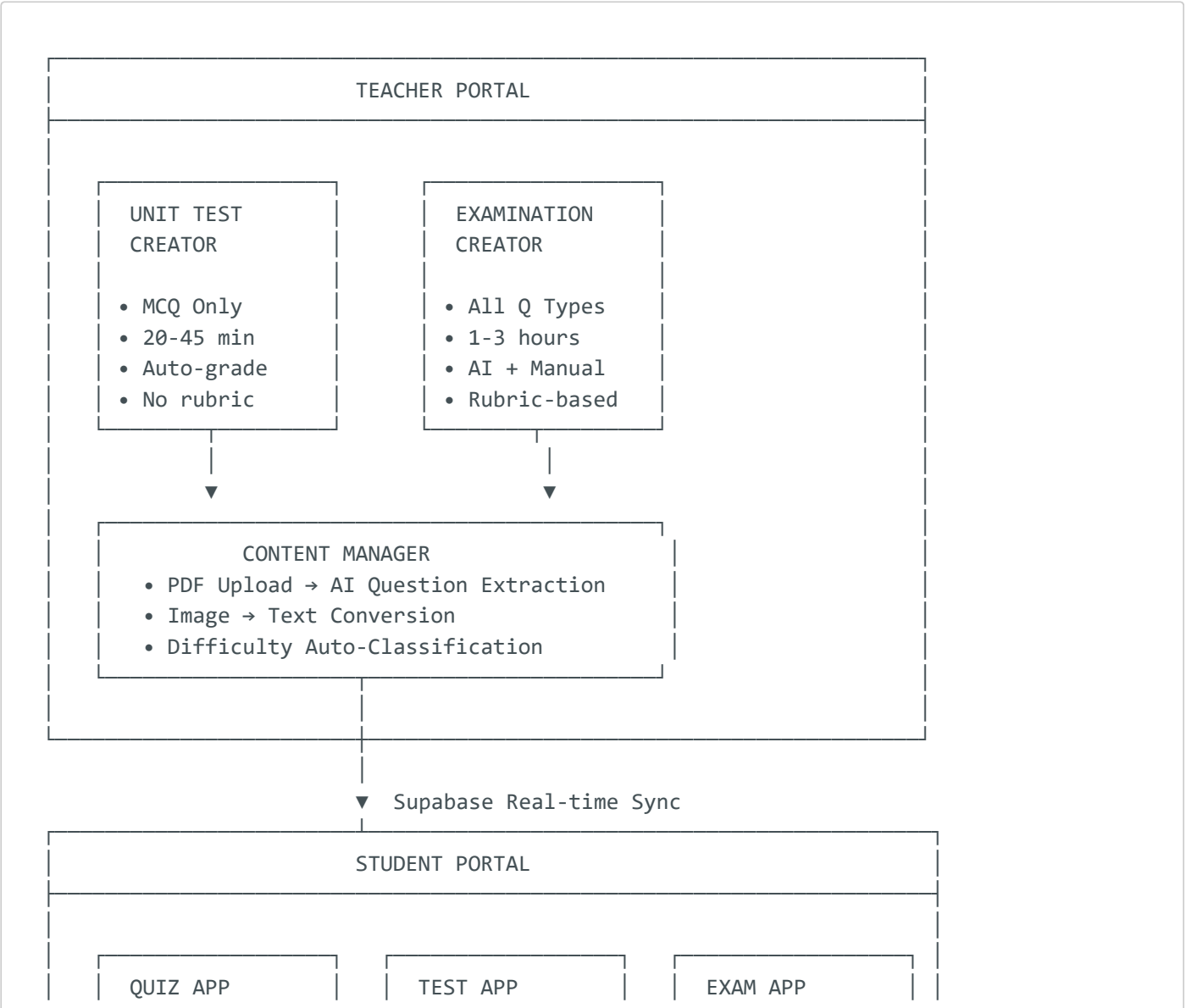
This section shows how features in the Teacher Portal directly connect to corresponding features in the Student Portal, ensuring a cohesive learning experience.

Teacher Creates	Student Experiences	Data Flow
Unit Test (Assessment Creator)	TestApp.tsx	Teacher defines questions → Student takes timed test → Auto-grade → Teacher reviews
Examination (Exam Creator)	ExaminationApp.tsx	Teacher creates exam + rubrics → Student takes secure exam → AI grades → Teacher approves
Quiz Questions (Content Manager)	QuizApp.tsx	Teacher uploads PDF → AI extracts questions → Student practices daily
Rubric Criteria (Grading Center)	Subjective Answer Feedback	Teacher defines rubric → AI uses for grading → Student sees detailed feedback

Teacher Creates	Student Experiences	Data Flow
Notifications (Broadcaster)	NotificationCenter.tsx	Teacher sends message → Student receives real-time alert
Performance Analytics	DashboardAnalysis.tsx	Teacher views class trends → Student sees personal insights
Personalized Feedback	Quiz/Test Results	Teacher approves AI feedback → Student reads encouragement
Intervention Alert	Motivation Messages	Teacher flags concern → Student receives compassionate nudge
Moderation Queue	Peer Review System	Teacher oversees flagged reviews → Student gives/receives anonymous feedback



Assessment Flow Diagram



- Daily practice
- Adaptive
- Hints enabled

- Chapter tests
- Basic timer
- Quick feedback

- Board-style
- Full lockdown
- Proctoring

DASHBOARD ANALYSIS

- Performance graphs
- AI recommendations
- Study streaks



Grading Flow Diagram

Student Submits Answer

Question Type?

MCQ

Subjective

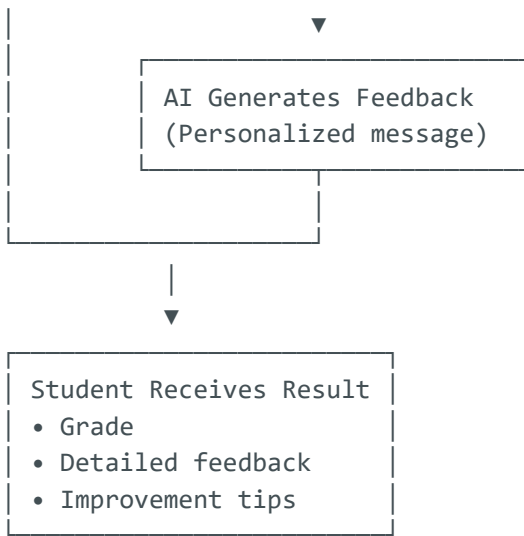
Auto-Graded
100%
Accurate

AI Grading with Rubric
(Gemini 2.5 Flash)

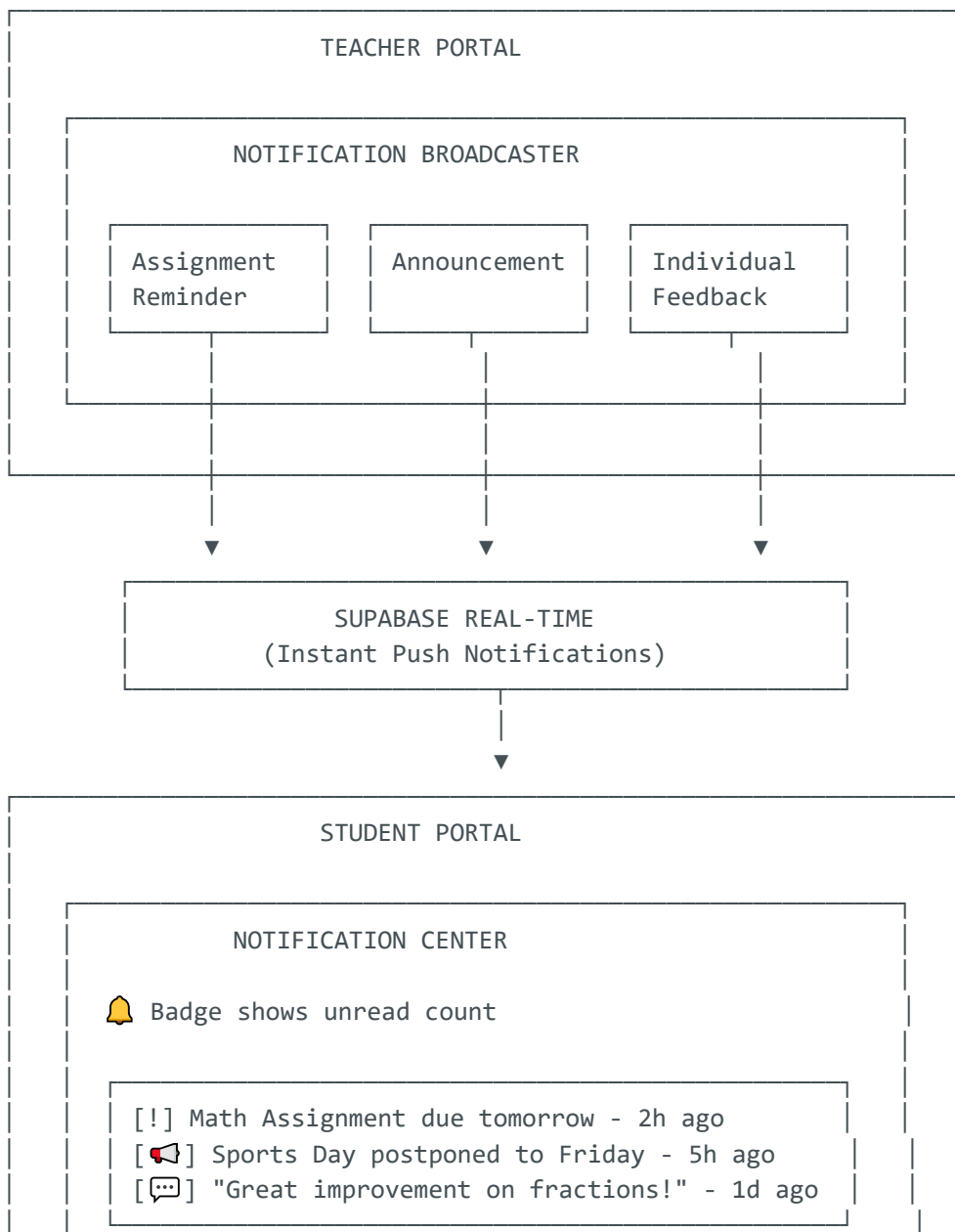
- Rubric criteria match
- Partial credit calc
- Confidence score

Teacher Review Queue

- View AI grade
- See confidence %
- Approve or Modify



Communication Flow Diagram





5. Enhanced Features & Missing Implementation Details

5.1 Additional Student Portal Features (Enhanced)

The following features are either enhanced versions of existing features or newly identified capabilities that strengthen the student learning experience:

Academic Enhancement Features

Feature	Description	Implementation Status	Component
Handwriting Converter	Convert handwritten notes to typed text using AI OCR	 Implemented	HandwritingConverter.tsx
Adaptive Quiz System	AI adjusts difficulty in real-time based on performance	 Implemented	AdaptiveQuiz.tsx
Peer Chat	Real-time messaging with classmates for study collaboration	 Implemented	PeerChat.tsx
Assessment Launcher	Unified entry point for all assessments	 Implemented	AssessmentLauncher.tsx

| **Social Learning Dashboard** | Leaderboards, study groups, peer tutoring |  Implemented | SocialLearningDashboard.tsx |

 Gamification & Engagement Features

Feature	Description	Target User	Status
Study Streaks	Track consecutive days of study activity	Students	 Active
XP Points System	Earn points for completing quizzes, tests, notes	Students	 Active
Weekly Leaderboards	Friendly competition among classmates	Students	 Active
Achievement Badges	Unlock badges for milestones (10 quizzes, 100% scores)	Students	 Active
Level Progression	Advance through learning levels (Beginner → Expert)	Students	 Active
Challenge Mode	Time-limited bonus challenges for extra XP	Students	 Planned

 AI-Powered Learning Enhancements

Feature	AI Model	Description	Trigger
Smart Study Planner	Lite	AI creates personalized study schedules based on exam dates and weak areas	Manual / Auto before exams
Topic Linking	Lite	Suggests related topics when studying (e.g., "Trigonometry" → "Geometry")	Automatic
Mistake Memory	Lite	Tracks repeated mistakes and resurfaces them in practice sessions	Automatic
Concept Maps	Flash	Generates visual mind maps from textbook content	On demand
Video Recommendations	Lite	Suggests YouTube/internal videos for difficult topics	After wrong answers

Feature	AI Model	Description	Trigger
Practice Problem Generator	Flash	Creates unlimited practice problems for any topic	On demand

 Writing Canvas Enhanced Features

Feature	Description	Technical Implementation
Infinite Pages	Scroll infinitely in any direction	Virtual viewport rendering
Paper Templates	Lined, grid, graph, dotted, blank, music staff	CSS background patterns
Color Picker	Full color wheel + recent colors	Custom component
Layer System	Multiple drawing layers for complex diagrams	Z-index management
Shape Tools	Rectangle, circle, arrow, line with snap-to-grid	Canvas geometry functions
Text Boxes	Type text directly on canvas	Absolute positioned divs
Image Insert	Paste or upload images into notes	Base64 encoding
PDF Export	Export pages as PDF with high quality	html2canvas + jsPDF
Cloud Sync	Auto-save to Supabase every 30 seconds	Real-time subscriptions
Offline Mode	Continue working without internet, sync when online	IndexedDB local storage

5.2 Additional School Portal Features (Enhanced)


OPTIONAL / FUTURE DEVELOPMENT — The following features are planned extensions for the School Portal. Development priority is on Teacher and Student portals.

 Administrative Dashboard Features

Feature	Description	Implementation Status
Real-time Occupancy	See which teachers are in class right now	 Active
Substitute Auto-Assignment	AI suggests best available substitute teacher	 Active
Parent Portal Integration	Parents can view child's dashboards	 Planned
Bulk Notification System	Send SMS/Email to all parents/teachers	 Active
Resource Booking	Book labs, auditoriums, projectors	 Planned
Maintenance Ticketing	Infrastructure repair requests	 Planned

Advanced School Analytics

Metric	Description	Visualization
School Performance Index	Aggregate score combining all academic metrics	Gauge chart
Teacher Load Balancing	Compare workload across teachers	Bar chart
Subject-wise Trends	Track improvement/decline by subject	Line chart
Attendance Heatmap	Visualize attendance patterns by day/month	Heatmap
Exam Score Distribution	Bell curve analysis per exam	Histogram
At-Risk Student Tracker	List of students with declining performance	Table + alerts

AI-Powered School Insights

Feature	AI Model	Output	Frequency
Dropout Prediction	Lite	Risk score per student (0-100)	Weekly
Teacher Performance Narrative	Flash	AI-generated evaluation report	Monthly

Feature	AI Model	Output	Frequency
Curriculum Coverage Tracker	Lite	% syllabus completed vs expected	Daily
Parent Communication Draft	Lite	Auto-generate PTM talking points	Before PTM
Board Exam Preparation Score	Flash	Class readiness for upcoming boards	Bi-weekly
Resource Utilization Report	Lite	Efficiency of lab/library usage	Monthly

5.3 Teacher-Student Parallel Feature Matrix

This matrix shows the complete parallel implementation between what teachers create and what students experience:

Teacher Action (Portal)	Student Experience (Portal)	Data Sync Method
Upload PDF to Content Manager	AI-generated questions appear in Quiz App	Supabase + Gemini
Create Unit Test	Test appears in Test App with timer	Supabase real-time
Create Examination with rubric	Exam appears in Examination App with lockdown	Supabase real-time
Define grading rubric	Subjective answers graded with rubric criteria	Stored in Supabase
Send notification	Notification appears in Notification Center	Supabase real-time
View class analytics	Personal insights appear in Dashboard Analysis	Parallel queries
Generate personalized feedback	Feedback appears after test submission	Gemini + Supabase

Teacher Action (Portal)	Student Experience (Portal)	Data Sync Method
Flag for intervention	Compassionate nudge message appears	Supabase trigger
Set assignment deadline	Assignment reminder in notification	Scheduled job
Mark answer as excellent	"Star Answer" showcase in leaderboard	Supabase flag

5.4 Implemented Features - Update Log

The following features have been recently implemented to complete the application:

✅ Critical Priority (IMPLEMENTED)

Feature	Portal	Status	Component
Exam Security	Teacher/School	✅ Implemented	<code>proctoringService.ts</code> , <code>ExaminationApp.tsx</code>
Parent Result Sharing Portal	School	✅ Implemented	<code>ParentPortal.tsx</code>
Offline Exam Mode	Student	✅ Implemented	<code>proctoringService.ts</code> (IndexedDB)
Batch Question Import	Teacher	✅ Implemented	<code>BatchQuestionImport.tsx</code>
Student Self-Assessment	Student	✅ Implemented	<code>SelfAssessment.tsx</code>

Proctoring Features Include:

- Anti-cheating measures (copy/paste prevention, tab switch detection, right-click blocking)
- Violation tracking with auto-submit after 3 violations
- Offline exam support with IndexedDB caching and sync

⚠️ **Honesty Note on Exam Security:** These proctoring measures are **browser-level deterrents**, not enterprise-grade lockdowns. They will discourage casual cheating

(switching tabs to Google, copy-pasting answers) but can be bypassed by a determined student using:

- A second device (phone, tablet) for looking up answers
- Virtual machines or screen-sharing software
- Browser developer tools (for advanced users)
- Screenshots via OS-level shortcuts

For high-stakes exams (Board Exams, entrance tests), these measures should be combined with **physical invigilation**. The proctoring system is best positioned as a "first line of deterrence" for regular classroom assessments, not a replacement for supervised examination environments.

Batch Import Features Include:

- CSV and Excel file parsing
- Template download for easy formatting
- Question validation with error reporting
- MCQ, short answer, and long answer support
- Preview and edit before finalizing import

✅ High Priority (IMPLEMENTED)

Feature	Portal	Status	Component
Peer Review System	Student		Implemented PeerReview.tsx
Teacher Collaboration	Teacher		Implemented TeacherCollaboration.tsx
Resource Sharing	Teacher		Via Collaboration TeacherCollaboration.tsx
Automated Report Cards	School		Implemented AutomatedReportCard.tsx

Peer Review Features:

- Anonymous review of peer answers
- 5-star rating system with constructive feedback
- Point rewards for participating
- Moderation guidelines
- Track reviews given and received

Teacher Collaboration Features:

- Create and manage question banks
- Visibility controls (private, school, public)

- Invite collaborators as editors or viewers
- Collaboration invitations system
- Star favorite banks

Automated Report Card Features:

- AI-powered insights (strengths, areas to improve)
- Subject-wise performance breakdown
- Attendance summary
- Progress trends visualization
- PDF export and print support
- Co-curricular activities and achievements

 Nice to Have (Future Roadmap)

Feature	Portal	Description	Priority
AR Lab Experiments	Student	Augmented reality for science experiments	Low
AI Tutor Avatar Student Animated character for chatbot Low			
Smart Desk IoT All Integrate with physical desk hardware sensors Low			

6. Competitive Analysis

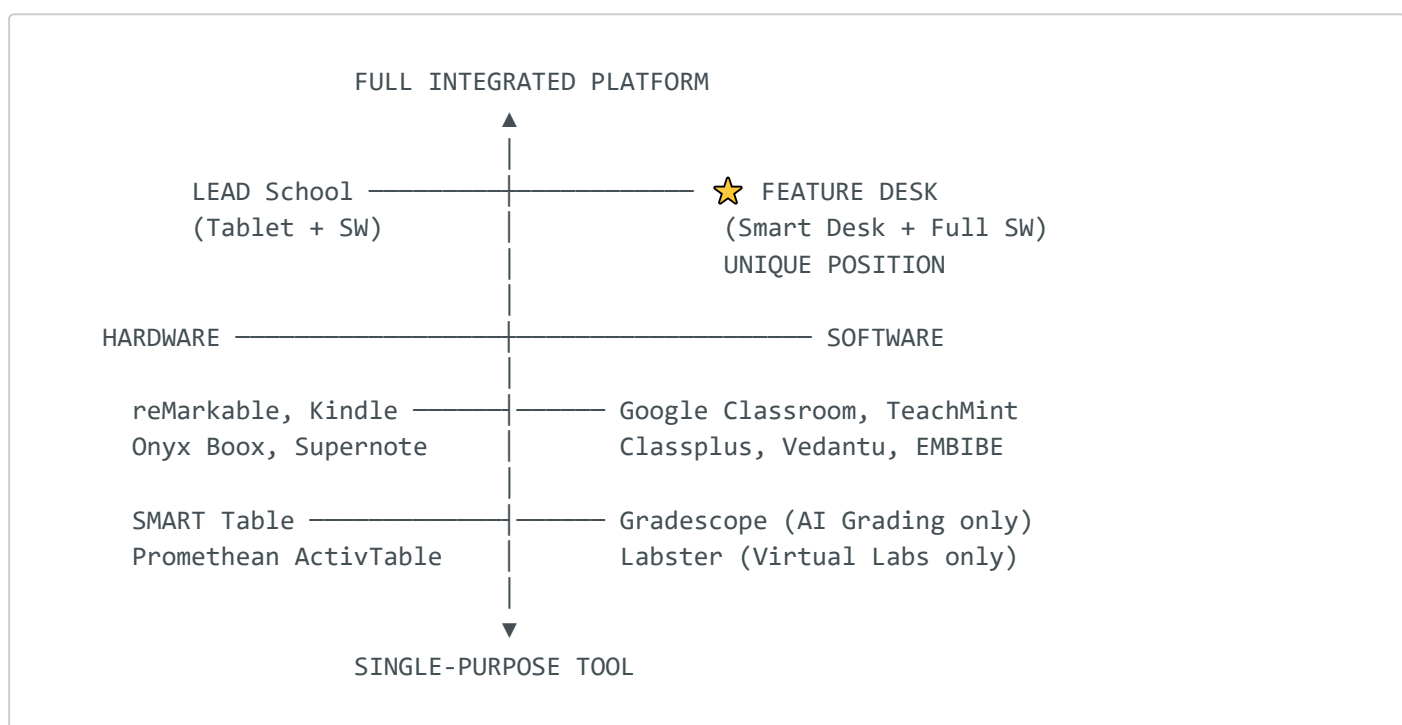
Feature Desk G occupies a **unique position** in the Indian EdTech market. Unlike tablet-based solutions that are easily lost, broken, or misused for entertainment, and unlike software-only platforms that depend on whatever hardware students happen to own, Feature Desk combines a **purpose-built smart desk** with a **full-stack AI-powered education platform**. This section analyzes direct and indirect competitors across hardware, software, and hybrid categories.

6.1 Feature Desk Key Differentiators

#	Differentiator	Why It Matters
1	Flip-Top Smart Desk	Only product that transforms from a normal writing desk to a digital touchscreen — no other competitor offers this

#	Differentiator	Why It Matters
2	Individual Student Desk	Unlike shared interactive tables (SMART/Promethean), each student has their own secure workspace for assessments
3	10 Core AI Capabilities (Gemini)	Deepest AI integration in any K-12 product — from question generation to rubric-based grading to Socratic hints
4	Triple-Layer Database	Supabase (structured) + Firebase (chat) + Cloudinary (media) ensures reliability at scale
5	₹50-100/month AI Cost	Gemini Flash/Lite pricing makes AI affordable for 1,000+ students — competitors charge ₹3,000-5,000/student/year
6	India-First Design	Built for CBSE/ICSE/State Board with 7 subjects including Tamil and Hindi
7	Teacher-Student Pipeline	Teacher uploads PDF → AI generates questions → Student takes adaptive quiz → AI grades → Teacher reviews — fully automated
8	Offline Exam Support	IndexedDB caching allows secure exams without internet, with auto-sync when reconnected

6.2 Market Positioning Map



6.3 Competitive Advantage Matrix

Feature	Feature Desk	LEAD Tab	reMarkable 2	Onyx Boox	SMART Table	Promethean
Flip-Top Smart Desk	 Unique					
Individual Student Desk		 (Tablet)	 (Tablet)	 (Tablet)	 (Shared)	 (Shared)
Normal + Digital Mode	 Unique					
Touchscreen + Stylus						
AI-Powered Learning (10 capabilities)						
Digital Writing Canvas (Infinite)		Partial				
Complete LMS Platform						Partial
School Mgmt (Attendance/Reports)						
Virtual Labs (Three.js/PhET)						
AI Grading (Rubric-based)						
Adaptive Learning		Partial				
Works without Internet						
India-focused (K-12, CBSE/ICSE)						
Affordable AI (₹50-100/1000 students)						
Teacher Collaboration						

Feature	Feature Desk	LEAD Tab	reMarkable 2	Onyx Boox	SMART Table	Promethean
(Question Banks)						
Peer Review System	✓	✗	✗	✗	✗	✗

6.4 Direct Competitors

Complete hardware + software educational solutions

#	Competitor	Type	Price Range	Key Features	Gap vs Feature Desk
1	LEAD School Tablet	Tablet + Software	₹15k-25k/student/yr	Pre-loaded curriculum, full LMS	✗ No flip-desk, just tablet
2	SMART Table	Interactive Table	\$5,000-15,000	Multi-touch collaborative surface	✗ Shared table, no individual desk
3	Promethean ActivTable	Interactive Table	\$8,000-12,000	46" touchscreen, built-in PC	✗ Shared table, no flip mechanism
4	BYJU'S FutureSchool Tab	Tablet + Content	₹30k-50k	Tablet with pre-loaded courses	✗ Consumer tablet, no desk
5	Lenovo Smart Classroom	Hardware + LMS	₹50k+/classroom	Interactive display, tablets, MDM	✗ Classroom-focused, not individual
6	HP Classroom Manager	Hardware + Software	Enterprise pricing	Device management, content	✗ No smart desk, just device fleet
7	Dell Latitude Education	Laptop/Tablet + LMS	₹40,000+	Ruggedized devices,	✗ Standard laptop, no

#	Competitor	Type	Price Range	Key Features	Gap vs Feature Desk
				classroom SW	dedicated desk

6.5 Indirect Competitors — Hardware

Digital writing devices that solve PART of the problem

A. E-Ink Writing Tablets (Standalone)

#	Product	Brand	Price	Features	Gap vs Feature Desk
1	reMarkable 2	reMarkable (Norway)	₹35k-40k	10.3" E-Ink, paper-like feel	✗ Just tablet, no desk, no LMS
2	reMarkable Paper Pro	reMarkable	₹55k-60k	Color E-Ink, improved writing	✗ No education platform, expensive
3	Kindle Scribe	Amazon	₹45k-50k	10.2" E-Ink, handwriting notes	✗ Reading-focused, no school features
4	Onyx Boox Tab Ultra C Pro	Onyx (China)	₹70k-90k	10.3" Color E-Ink, Android	✗ General purpose, no integrated LMS
5	Supernote A5X	Ratta (China)	₹35k-50k	E-Ink, distraction-free	✗ Just a tablet, no education focus

B. Smart Pens & Digital Writing Systems

#	Product	Brand	Price	Features	Gap vs Feature Desk
1	Unlox (Neo Smartpen)	NeoLAB	₹15k-25k	Smart pen + special paper	✗ Requires special paper, no touchscreen
2	Wacom Intuos Pro	Wacom	₹25k-40k	Professional drawing tablet	✗ Graphics tablet, not integrated desk

#	Product	Brand	Price	Features	Gap vs Feature Desk
3	XP-Pen Artist 24 Pro	XP-Pen	₹50k-70k	24" pen display, 2K resolution	✗ Art/design focus, no education SW

6.6 Indirect Competitors — Software Only

Platforms that need separate hardware

#	Software	Focus	Price	Target	Gap vs Feature Desk
1	LEAD School LMS	Complete K-12 LMS	₹3,000-5,000/yr	Schools	✗ Needs separate tablet
2	TeachMint	LMS + ERP	₹200-500/student/yr	Schools	✗ No hardware, no digital canvas
3	Classplus	Coaching management	₹5,000-15,000/yr	Tutors	✗ Coaching-focused, no assessment AI
4	Vedantu	Live tutoring	₹15,000-40,000/yr	Students	✗ Video classes only, no writing tools
5	Toppr	Practice + Tests	₹10,000-25,000/yr	Students	✗ MCQ-only, no subjective AI grading
6	Extramarks	Content + LMS	₹5,000-15,000/yr	Schools	✗ Content delivery, no creation tools
7	EMBIBE	AI Learning Platform	₹3,000-5,000/yr	Students	✗ No writing canvas, no desk
8	Gradescope	AI Grading	\$3-5/student/course	Universities	✗ Only grading, no full platform
9	Google Classroom	Free LMS	Free	All	✗ No writing, no AI, no desk

#	Software	Focus	Price	Target	Gap vs Feature Desk
10	Labster	Virtual Labs	\$3,000+/class	Labs	✗ Standalone, very expensive

6.7 Classroom Infrastructure (India)

C. Interactive Classroom Hardware

#	Product	Brand	Price	Features	Gap vs Feature Desk
1	Promethean ActivPanel	Promethean	₹3-8 lakhs	65-86" interactive display	✗ Classroom display, not student desk
2	SMART Board MX Series	SMART Technologies	₹4-10 lakhs	Interactive whiteboard	✗ Wall-mounted, not individual desk
3	ViewSonic ViewBoard	ViewSonic	₹2-6 lakhs	Interactive flat panel, Android	✗ Classroom presentation tool

D. Digital Classroom Furniture (India)

#	Provider	Type	Features	Gap vs Feature Desk
1	Edutek Smart Class	Digital Classroom	Interactive panels, projectors	✗ No smart desk, just displays
2	Roombr	Digital Classroom	Complete classroom solutions	✗ No flip-desk hardware
3	Godrej Interio	Traditional Furniture	Ergonomic desks, chairs	✗ No digital features
4	Nilkamal Edge	School Furniture	Student desks, benches	✗ Purely physical furniture

6.8 Price Comparison Summary

Solution Type	Product	Annual Cost (per student)	What You Get
Feature Desk G	Smart Desk + Full Platform	Hardware + ~₹15-20/student/month	Desk + LMS + 10 Core AI Engines + Canvas + Labs
Tablet + LMS	LEAD School	₹15,000-25,000	Tablet + curriculum + basic LMS
Software Only	TeachMint	₹200-500	LMS + ERP (no hardware, no AI)
Software Only	Vedantu	₹15,000-40,000	Live video tutoring only
Software Only	Toppr	₹10,000-25,000	Practice MCQs + test series
E-Ink Tablet	reMarkable 2	₹35,000-40,000 (one-time)	Writing tablet only, no LMS
Shared Table	SMART Table	\$5,000-15,000 (one-time)	Shared surface, no individual use
AI Grading	Gradescope	\$3-5/student/course	Grading only, manual upload

6.9 Strategic Competitor Analysis & Threat Assessment

Deep dive into why existing solutions fail to solve the core classroom problem.

1. Analysis of Direct Competitors (Hardware + Software)

- **LEAD School & Tablet Ecosystems:** The primary weakness of tablet-based solutions (LEAD, BYJU'S) is **ergonomics and distraction**. A tablet on a flat desk encourages poor posture and is easily dropped. Feature Desk's "integrated flip-top" design solves this by providing a stable, angled surface that is robust enough for classroom wear-and-tear. Furthermore, tablets often become consumption devices (watching videos), whereas the Feature Desk is engineered as a creation workstation.
- **Interactive Tables (Smart/Promethean):** These products are designed for **collaboration**, not **individual assessment**. You cannot administer a secure Board

Exam on a shared Smart Table. Feature Desk provides the privacy and security needed for individual testing, which is critical for the Indian education system where individual performance metrics are paramount.

2. Analysis of Indirect Competitors (Software Only)

- **LMS Platforms (Google/TeachMint):** These platforms excel at **logistics** (attendance, fees, file sharing) but fail at **pedagogy**. They do not "understand" the content. Feature Desk's Gemini AI doesn't just store a PDF; it reads it, generates questions, and grades answers. We move the value proposition from "School Management" to "Instructional Improvement".
- **AI Graders (Gradescope):** While powerful, they are **disconnected**. A teacher has to scan papers, upload them, grade them, and then manually transfer grades to an LMS. Feature Desk integrates this flow: the student writes on the desk, the AI grades it instantly, and the LMS is updated automatically—saving 100% of the administrative overhead key in teacher burnout.
- **Indian EdTech (Vedantu/Toppr/Extramarks):** These focus on **content consumption** — video lessons, pre-made quizzes, and test series. They don't support teachers creating their own assessments, don't offer AI-assisted grading of handwritten answers, and provide no collaborative features between teachers. Feature Desk is a **creation platform**, not a content library.

3. Analysis of Indirect Competitors (Hardware Devices)

- **E-Ink Tablets (reMarkable/Kindle):** These provide a superior writing feel but are **functionally limited**. They lack color (critical for Biology/Chemistry diagrams), cannot run interactive simulations (Virtual Labs requires WebGL), and have no video streaming capabilities (for our Live Code Studio). Feature Desk matches the natural writing experience via stylus but retains the versatility of a full color touchscreen for rich media.

6.10 Competitive Moat Summary

Feature Desk G's defensible advantage rests on **three pillars** that no single competitor currently replicates:



HARDWARE MOAT	AI MOAT	ECOSYSTEM MOAT
• Flip-top desk form factor • Individual not shared workspace • Normal + Digital dual mode • Stylus-first design	• 10 Gemini AI features • Content-aware (reads + grades) • ₹50-100/month for 1000 students • Auto key rotation for reliability	• Teacher ↔ Student content pipeline • Question bank sharing (collab) • Peer review ecosystem • 18-table Supabase schema

Bottom Line: Competitors must replicate the **hardware** (requires manufacturing), the **AI depth** (requires deep Gemini integration across 10 core capabilities), AND the **ecosystem** (requires building Teacher-Student data pipelines) — solving all three simultaneously is Feature Desk's structural advantage.

7. Software Implementation & Architecture

7.1 AI Model Strategy: "The Dual-Brain Architecture"

Feature Desk G uses a cost-optimized "Dual-Brain" approach using Google's Gemini models. This ensures high intelligence for complex tasks while maintaining speed and low cost for routine operations.

Task Category	AI Model Used	Reasoning
Complex Reasoning	Gemini 2.5 Flash	Used for heavy cognitive tasks like generating exams from PDFs, grading complex biology diagrams, and logic-based math grading. Needs high context window and reasoning capabilities.
Multimodal Vision	Gemini 2.5 Flash	Used for "Image-to-Quiz"; analyzing handwritten teacher notes or printed worksheets to convert them into digital assessments.
High Frequency / Low Latency	Gemini 2.5 Flash-Lite	Used for real-time tasks like "Mistake Pattern Detection", summarizing student feedback, generating parent emails, and class readiness forecasting. Optimized for speed and cost.

7.2 Database Architecture: "Hybrid Memory"

The platform leverages a hybrid database structure to handle real-time classroom data vs. large-scale analytics.

Primary Database: Supabase (PostgreSQL)

- **Role:** The "Source of Truth" for structured data.
- **Key Usage:**
 - **Authentication:** Secure login for Students/Teachers/Admins.
 - **Real-time Sync:** Synchronizing the "Live Code Studio" so students see code updates instantly.
 - **Relational Data:** Storing **Students**, **Teachers**, **Classes**, and **Schedules**.
 - **Row Level Security (RLS):** Ensuring a student can only see their own grades.

Secondary Database: Firebase (Firestore Only)

- **Role:** Real-time chat data storage.
- **Key Usage:**
 - **Firestore:** Stores **chat_sessions** and **chat_messages** for the AI Chatbot.

Media Storage: Cloudinary

- **Role:** All media file uploads (PDFs, images, documents, canvas JSON data).
- **Key Usage:**
 - **Canvas JSON:** Stores Writing Canvas stroke data as raw JSON blobs.
 - **Teacher PDFs:** Uploaded PDFs for content AI pipeline.
 - **Exam Answer Sheets:** Student handwritten answer images.
 - **Documents:** Word/Excel/PPT files uploaded by teachers.

7.3 Implementation Roadmap (Phases)

MVP Features (Weeks 1-4)

Priority Tier 1: Core Learning & Teaching

1.  **Socratic Hint System** (Student) - Prevents Cheating, Builds Critical Thinking
2.  **Intelligent Question Generation** (Teacher) - Saves 10+ Hours/Week
3.  **Rubric-Based AI Grading** (Teacher) - Exam Support at Scale
4.  **Auto Lesson Plan Generation** (Management) - Handles Absences Seamlessly

5.  **Mistake Pattern Detection (Teacher)** - Identifies Class-Wide Gaps

Phase 2 (Q2 2026) - Enhanced Learning

- Adaptive Reinforcement Quiz Generator
- Concept Explanation on Demand
- Context-Aware Chatbot
- Personalized Feedback Generation
- Student Readiness Forecast

Phase 3 (Q3 2026) - Advanced Analytics

- Learning Velocity Narrative
- Confidence-Based Learning Insights
- Predictive Dropout Risk Analysis
- Benchmarking Insight Report
- Question Effectiveness Analysis

Phase 4 (Q4 2026) - Engagement & Personalization

- Achievement Narrative Generation
- Personalized Motivation Messages
- Learning Style Detection
- Scenario Consequence Generator
- Weekly Progress Narratives

7.4 Cost Analysis & ROI

Assumptions:

- 500 quiz sessions/day (avg 2-3 interactions per student)
- 50 question generations/week
- 200 gradings/day
- 1 substitute lesson plan/day

Token Breakdown (API Costs Only):

- **Gemini 2.5 Flash:** ~2M tokens/month = ~\$0.50
- **Gemini 2.5 Flash Lite:** ~5M tokens/month = ~\$0.10
- **AI API Subtotal:** ~₹50-₹100/month for 1,000 students.

Full Total Cost of Ownership (TCO) per School:

The AI token cost above is only one component. A realistic monthly operating cost includes:

Service	Free Tier Allowance	Estimated Monthly Cost (1,000 Students)
Gemini AI API	Free tier covers ~2M requests	₹50-100
Supabase (DB + Auth + Realtime)	500MB DB, 50K auth users	₹0 (free tier) → ₹1,500-3,000 at scale
Firebase Firestore (Chat)	1GB storage, 50K reads/day	₹0 (free tier) → ₹500-1,500 at scale
Cloudinary (Media/PDF storage)	25GB bandwidth, 25K transforms	₹0 (free tier) → ₹1,000-2,500 at scale
Hosting (Vercel/Netlify)	Free tier for hobby	₹0 → ₹1,000-2,000 (pro)
Domain + SSL	—	₹500-1,000/year
Technical Support/Maintenance	—	₹5,000-10,000
Total Estimated TCO	—	₹8,000-20,000/month at scale

Note: Free tiers cover most usage for pilot schools (< 200 students). At scale (1,000+ students), costs increase but remain significantly lower than competitors charging ₹3,000-5,000/student/year.

ROI: Teachers save 10+ hours/week, Students learn better, Schools run efficiently. The total platform cost at scale (₹15,000-20,000/month) translates to roughly **₹15-20 per student/month** — still a fraction of competitor pricing.

8. Installation & Setup

Prerequisites

- Node.js (v19+)
- npm or yarn

- Supabase account (PostgreSQL - structured data)
- Firebase project (Firestore - chat data only)
- Cloudinary account (media file uploads - pre-configured in code)
- Google Gemini API Key

Steps

1. Clone the Repository:

```
git clone [repository-url]
cd Feature_Desk_G
```

2. Install Dependencies:

```
npm install
```

3. Environment Variables: Create a .env file in the root directory:

```
# Supabase Configuration (PostgreSQL)
VITE_SUPABASE_URL=your_supabase_url
VITE_SUPABASE_ANON_KEY=your_supabase_key

# Gemini AI Configuration
VITE_GEMINI_API_KEY=your_gemini_api_key

# Firebase Configuration (Firestore for chat only)
VITE_FIREBASE_API_KEY=your_firebase_key
VITE_FIREBASE_AUTH_DOMAIN=your_project.firebaseio.com
VITE_FIREBASE_PROJECT_ID=your_project_id
VITE_FIREBASE_STORAGE_BUCKET=your_project.appspot.com
VITE_FIREBASE_MESSAGING_SENDER_ID=your_sender_id

# Optional Feature Flags
VITE_DEBUG_MODE=false
VITE_ENABLE_AI_GRADING=true
VITE_ENABLE_QUESTION_GENERATION=true
```

Note: Cloudinary is pre-configured in the codebase (`cloudinaryService.ts`) with cloud name `dbmrflbfg` and unsigned upload preset `feature_desk_unsigned`. No environment variable needed.

4. Run Development Server:

```
npm run dev
```

8.5 Application Routes (App.tsx)

The application uses React Router v7 with role-based route guards:

Student Routes (requires `userType === 'student'`)

Route	Component	Description
<code>/</code>	<code>StudentDashboard</code>	Main dashboard with WritingCanvas + CentralHub
<code>/quiz</code>	<code>QuizApp</code>	Adaptive quiz with AI-powered questions
<code>/test</code>	<code>TestApp</code>	Unit Test interface (no password required)
<code>/exam</code>	<code>ExaminationApp</code>	Secure proctored examination with lockdown
<code>/notes</code>	<code>NotesApp</code>	Note-taking and management
<code>/dashboard</code>	<code>DashboardAnalysis</code>	Performance analytics and badges
<code>/chatbot</code>	<code>LiveChatbot</code>	AI chatbot with session management
<code>/history</code>	<code>HistoryViewer</code>	Past quiz/exam result history
<code>/notifications</code>	<code>NotificationCenter</code>	Teacher broadcasts and alerts
<code>/life-activity</code>	<code>LifeActivityApp</code>	Daily activity tracking
<code>/social-learning</code>	<code>SocialLearningDashboard</code>	Peer tutoring and leaderboards
<code>/self-assessment</code>	<code>SelfAssessment</code>	Self-evaluation with AI suggestions
<code>/peer-review</code>	<code>PeerReview</code>	Anonymous peer feedback system

Teacher Routes (requires userType === 'teacher')

Route	Component	Description
/	TeacherDashboard	Dashboard with all teacher modules
/chatbot	LiveChatbot	AI chatbot (shared component)

School Admin Routes (requires userType === 'school')

Route	Component	Description
/	SchoolDashboard	School-wide metrics and management

Shared Components (always mounted)

- FloatingAIChatbot — A floating bubble available on all student and teacher pages for quick AI access.

8.6 Core Library Files (src/lib/)

File	Purpose	Key Functions
gemini.ts (575 lines)	Gemini AI core — API key rotation, dual-model exports, student AI features	callWithFallback(), convertHandwritingToText(), generateAdaptiveQuiz(), generateSocraticHints(), generateConceptExplanation(), analyzeConfidencePatterns(), generateWeeklyNarrative()

File	Purpose	Key Functions
<code>teacherAI.ts</code> (773 lines)	Teacher AI features — Question generation, grading, analytics	<code>generateQuestionsFromPDF()</code> , <code>extractQuestionsFromImage()</code> , <code>gradeAnswerWithRubric()</code> , <code>gradeImageAnswerWithRubric()</code> , <code>calculatePartialCredit()</code> , <code>detectMistakePatterns()</code> , <code>forecastClassReadiness()</code> , <code>chatWithDocument()</code>
<code>contentAI.ts</code> (483 lines)	Content pipeline — Upload → Process → Generate Questions + Learning Methods	<code>processAndGenerateContent()</code> , <code>generateQuestionsFromContent()</code> , <code>generateLearningMethodFromContent()</code> , <code>generateAdaptiveQuizFromMaterials()</code> , <code>extractTextFromPDF()</code> , <code>gradeAnswerAgainstContent()</code>
<code>schoolAI.ts</code> (392 lines)	School AI features — Lesson plans, dropout analysis, benchmarks	<code>generateLessonPlan()</code> , <code>generateAbsentDayNotification()</code> , <code>generateAnnualReport()</code> , <code>analyzeDropoutRisk()</code> , <code>generateBenchmarkReport()</code> , <code>analyzeExamReadiness()</code>
<code>teacherDb.ts</code> (735 lines)	Teacher database layer — CRUD for students, assessments, grading, analytics	<code>getStudentsByClass()</code> , <code>createAssessment()</code> , <code>getPendingResults()</code> , <code>approveGrade()</code> , <code>getClassAnalytics()</code> , <code>broadcastNotification()</code>
<code>schoolDb.ts</code> (413 lines)	School database layer — Stats, absences, schedules, risk analysis	<code>getSchoolStats()</code> , <code>getTeacherAbsences()</code> , <code>assignSubstitute()</code> , <code>getClassPerformance()</code> , <code>getStudentAttendanceRisk()</code>
<code>questionDb.ts</code> (503 lines)	Question bank system — Stores AI-generated questions and learning methods	<code>saveTeachingMaterial()</code> , <code>saveGeneratedQuestions()</code> , <code>getRandomQuizQuestions()</code> , <code>getQuestionStats()</code> , <code>saveLearningMethod()</code>

File	Purpose	Key Functions
<code>db.ts</code> (307 lines)	Hybrid save helpers — Orchestrates Supabase + Cloudinary saves	<code>saveQuizResultHybrid()</code> , <code>saveExamSubmissionHybrid()</code> , <code>saveCanvasNoteHybrid()</code>
<code>cloudinaryService.ts</code> (203 lines)	Cloud media uploads — File/PDF/Doc/JSON uploads	<code>uploadFile()</code> , <code>uploadPdf()</code> , <code>uploadDocument()</code> , <code>uploadJson()</code> , <code>_getResourceType()</code>
<code>firebaseService.ts</code> (183 lines)	Firestore chat service — Session-aware chat persistence	<code>createChatSession()</code> , <code>saveChatMessage()</code> , <code>getChatHistory()</code> , <code>subscribeToChat()</code>
<code>proctoringService.ts</code> (402 lines)	Exam security — Anti-cheat, screenshots, offline support	<code>captureScreenshot()</code> , <code>setupAntiCheating()</code> , <code>startPeriodicScreenshots()</code> , <code>saveExamForOffline()</code> , <code>syncOfflineData()</code>
<code>supabase.ts</code>	Supabase client — Single instance configuration	<code>supabase</code> client export
<code>firebase.ts</code>	Firestore client — App and Firestore initialization	<code>db</code> Firestore instance export

8.7 Key Undocumented Component: ExamAnswerInput

ExamAnswerInput (`ExamAnswerInput.tsx` — 56KB, 1158 lines)

A full-featured drawing canvas embedded inside examinations for handwritten answer submission:

- **Dual Input Mode:** Students can switch between **write** (drawing canvas) and **type** (text input) per question
- **Drawing Tools:** Pen, Highlighter, Eraser, Shapes (Rectangle, Circle, Line, Arrow, Star, Heart, Hexagon)
- **Canvas Features:** Zoom (0.5x–2.0x), Pan, Undo/Redo with history stack, Color palette (9 colors)
- **Paper Styles:** Blank, Ruled, Grid, Dotted backgrounds
- **Virtual Keyboard:** Draggable, resizable on-screen keyboard with symbol support
- **Auto-Save:** Canvas data auto-captured and encoded as base64 on visibility change / page unload
- **Security:** Hooks into **proctoringService** for visibility change detection during exams
- **Responsive:** Adapts to different screen sizes with virtual width normalization (900px)

9. Database Setup & Schema

9.1 Supabase (PostgreSQL) Setup

Database Files Location

```
database/
├── supabase_schema.sql      # Complete schema (17KB, 400+ lines)
├── demo_data_class7.sql    # Demo data for 100 students (39KB, 520+ lines)
└── README.md               # Setup instructions
```

Complete Table Schema

Table	Description	Key Columns
classes	School classes 1-12	id, class_name, section, academic_year
subjects	Subject master (7 subjects)	code, subject_name, icon_emoji, color
teachers	Teacher records	email, teacher_name, assigned_class, assigned_subjects

Table	Description	Key Columns
students	Student records (100 demo)	roll_number, student_name, current_class, password
student_subjects	Student-Subject enrollment	student_id, subject_code
assessments	Quizzes, tests, exams	title, exam_type, questions (JSONB), total_marks
quiz_results	Student scores	score, grade, percentage (auto-calculated), teacher_approved
quiz_responses	Detailed per-question responses	question_id, student_answer, is_correct, marks
student_notes	Digital notebook entries	title, content, subject_code, note_type, tags
notifications	Teacher-to-student messages	title, message, notification_type, priority
student_attendance	Daily attendance records	status (present/absent/late), attendance_date
leaderboard	Points & gamification	total_points, current_rank, badges, level, streak_days
peer_help_requests	Study help requests	topic, status, requester_id, helper_id
peer_sessions	Help session records	rating, feedback, duration
peer_messages	Session chat messages	content, message_type, sender_id
student_interactions	Activity tracking	interaction_type, details (JSONB), duration
study_materials	Uploaded teaching materials	file_url, ai_summary, key_topics

Supported Subjects (7 Total)

Code	Name	Icon	Color
MATH	Mathematics		#3B82F6 (Blue)

Code	Name	Icon	Color
SCIENCE	Science		#10B981 (Green)
ENGLISH	English		#8B5CF6 (Purple)
HINDI	Hindi (हिंदी)		#F59E0B (Amber)
TAMIL	Tamil (தமிழ்)		#DC2626 (Red)
SOCIAL	Social Studies		#EF4444 (Red)
COMPUTER	Computer Science		#06B6D4 (Cyan)

Running Supabase Schema

1. Go to **Supabase Dashboard** → **SQL Editor**
2. Create new query → Paste contents of `database/supabase_schema.sql`
3. Click **Run** (creates tables, indexes, RLS policies)
4. Create new query → Paste contents of `database/demo_data_class7.sql`
5. Click **Run** (inserts 100 students + assessments + results)

9.2 Firebase Setup (Firestore Only)

Firestore Collections

Collection	Role	Document Structure
<code>chat_sessions</code>	Stores active chat threads	<code>{ userId, title, createdAt, lastMessageAt }</code>
<code>chat_history</code>	Stores individual messages	<code>{ userId, sessionId, content, role, timestamp }</code>

Note: Firebase is used **only** for Firestore (chat data). All media file uploads go through **Cloudinary**.

9.3 Cloudinary Setup (Media Storage)

Service	Configuration	Details
Cloud Name	<code>dbmrflbfg</code>	Configured in <code>cloudinaryService.ts</code>

Service	Configuration	Details
Upload Preset	feature_desk_unsigned	Unsigned preset for client-side uploads

Upload Endpoints

Method	Resource Type	Use Case
uploadFile()	auto-detected	General file uploads (images, etc.)
uploadPdf()	raw	Teacher PDF uploads
uploadDocument()	raw	Word, Excel, PPT documents
uploadJson()	raw	Canvas stroke data as JSON blobs

9.4 Demo Data (Class 7)

Student Data (100 Students)

- **Roll Numbers:** 7A001 to 7A100
- **Names:** Realistic Indian names (Aarav Sharma, Aanya Patel, etc.)
- **Default Password:** 123456
- **Current Class:** 7 (Class 7-A)

Assessment Data (16 Assessments)

Subject	Assessments	Types
Mathematics	3	Unit Test, Quiz, Mid-Term
Science	3	Unit Test, Quiz, Mid-Term
English	2	Grammar Test, Vocabulary Quiz
Hindi	2	व्याकरण परीक्षा, Weekly Quiz
Tamil	2	இலக்கணம் Test, வினாடி வினா
Social Studies	2	History Test, Geography Quiz
Computer	2	Programming Test, Hardware Quiz

Quiz Results

- ~1200+ results across all students and assessments

- Realistic score distribution (15-95%)
- Grades: A+, A, B, C, D

Additional Demo Data

- 30 days of attendance (weekdays only)
- 700 leaderboard entries (all students × all subjects)
- 50+ peer help requests
- 200+ student notes
- 5 class notifications

9.5 Demo Login Credentials

Role	Credentials	Access
Teacher	Email:teacher@demo.com, Password: teacher123	Full teacher dashboard
Student	Roll:7A001-7A100, Password: 123456	Individual student portal
School Admin	Email:admin@school.edu, Password: admin123	School management dashboard

Sample Students:

Roll	Name	Gender
7A001	Aarav Sharma	Male
7A002	Aanya Patel	Female
7A010	Diya Nair	Female
7A025	Sahil Chatterjee	Male
7A050	Gauri Patil	Female
7A100	Nandini Sen	Female

10. Recent Updates (February 2026)

10.1 Database Infrastructure

- ☒ Complete Supabase schema with 15+ tables
- ☒ Row Level Security (RLS) policies
- ☒ Auto-calculated percentage field in quiz_results
- ☒ Performance-optimized indexes

10.2 Demo Data Generation

- ☒ 100 Class 7 students with realistic Indian names
- ☒ 16 assessments across 7 subjects
- ☒ 1200+ quiz results with varied scores
- ☒ 30 days of attendance records
- ☒ Leaderboard data for all students

10.3 Tamil Language Support

- ☒ Added Tamil as 7th subject
- ☒ Created Tamil assessments with proper Unicode
- ☒ Tamil peer help topics
- ☒ Updated teacher assignments

10.4 Code Quality & Architecture

- ☒ Updated AuthContext with Class 7 demo data
- ☒ Removed hardcoded Class 10 references
- ☒ Hybrid database helper functions (`db.ts: saveQuizResultHybrid, saveExamSubmissionHybrid, saveCanvasNoteHybrid`)
- ☒ Cloudinary media storage integration (replaced MongoDB/Firebase Storage)
- ☒ API Key Fallback System with automatic key rotation
- ☒ Content AI pipeline (`contentAI.ts`) for material → questions → learning methods
- ☒ Question Bank system (`questionDb.ts`) for AI-generated question storage
- ☒ ExamAnswerInput component with full drawing canvas for handwritten exam answers
- ☒ Self-Assessment and Peer Review routes added

10.5 Newly Documented Components

Teacher Portal (4 components added):

-  **TeacherCollaboration.tsx** (43KB) — Cross-teacher question bank sharing with roles
-  **BatchQuestionImport.tsx** (25KB) — Bulk Excel/CSV question upload with validation
-  **AssessmentManager.tsx** (31KB) — CRUD for existing assessments with filtering
-  **StudyMaterialsManager.tsx** (34KB) — File organizer with Cloudinary upload and AI summaries

Student Portal (4 components added):

-  **AssessmentLauncher.tsx** (13KB) — Unified quiz/test/exam launch screen
-  **HandwritingConverter.tsx** (6KB) — Gemini Vision OCR for canvas-to-text
-  **PeerReview.tsx** (32KB) — Anonymous peer feedback with star ratings
-  **AdaptiveQuiz.tsx** (7KB) — Rapid-fire micro-learning quiz sessions

10.6 Documentation Updates

-  School Portal marked as optional/future development
-  Teacher Dashboard tabs updated to reflect all 10 actual tabs
-  ER Diagram added (18 tables with relationships)
-  Triple-Layer Data Flow Diagram added (Supabase/Firebase/Cloudinary)
-  Authentication Flow Diagram added (3 user types)
-  AI Content Pipeline Flow Diagram added (Upload → Generate → Deliver)
-  All MongoDB and Firebase Storage references removed
-  Teacher AI Feature #28 (Interactive Document Chat) added
-  Peer Review added to Producer-Consumer mapping

11. Deployment

Live URL

Production: <https://featuredeskx.netlify.app>

Build Commands

```
# Development
npm run dev

# Production Build
npm run build

# Deploy to Netlify
netlify deploy --prod --dir=dist
```

Build Output

```
dist/
├─ index.html           (~1.5 KB)
├─ assets/
│   ├─ index.css        (~72 KB, gzipped: 11 KB)
│   ├─ index.js         (~1.1 MB, gzipped: 320 KB)
│   ├─ xlsx.js          (~430 KB, gzipped: 143 KB)
│   └─ ...other chunks
```

Documentation generated by specific user request. Last Updated: February 13, 2026 — 4:00 PM IST

Primary Focus: Student Portal (16 Apps) + Teacher Portal (10 Tabs) | **Optional:** School Portal (Future Roadmap)